# 1 HI-Jass User Manual

## 1.1 Running the application

### 1.1.1 Linux

```
./run_hi_jass.sh
```

### 1.1.2 Windows

Double-click `run_hi_jass.bat`, or run:

```
.\run_hi_jass.bat
```

The launcher creates the virtual environment, installs the dependencies, and starts the GUI.

## 1.2 Default NBI configuration

The default configuration contains two neutral beams:

| Beam | Species | Injection energy | Injected power | Tangent $R_t$ | Direction |
|---|---|---|---|---|---|
| NBI-1 | D | 120 keV | 10 MW | 0.5 m | co |
| NBI-2 | T | 180 keV | 0.1 MW | 0.6 m | counter |

## 1.3 Beam vertical angle

`BeamParams.vertical_angle_deg` ($\alpha$, HOT-Jass: NBI "vertical angle", default 0) inclines the beam axis to its horizontal projection about the fixed tangency point $(R_t, Z_t)$; $\alpha > 0$ rises along the travel. With $y$ the horizontal distance from the tangency point (negative before it):

$$R(y) = \sqrt{R_t^2 + y^2}, \qquad Z(y) = Z_t + y \tan\alpha, \qquad ds = dy/\cos\alpha.$$

`physics.tangential_chords` finds the entry and exit through the LCFS by bisection on the outermost crossings (samples in between outside the plasma, e.g. in the hole, carry $n_e = 0$); the centre-post blocks at the same $y$ as for $\alpha = 0$. Shine-through, the birth profile and the first-orbit-loss chord all use this path, and the vertical ray offsets of a finite beam become $v/\cos\alpha$ in $Z$. The birth pitch (below) includes the inclination, and the NBI torque is multiplied by $\cos\alpha$, since only the horizontal velocity carries toroidal momentum. $\alpha = 0$ keeps the horizontal-chord geometry exactly.

## 1.4 Field directions and the beam birth pitch

Convention: $I_p > 0$ and $B_t > 0$ both counter-clockwise seen from above ($+\phi$ in right-handed $(R, \phi, Z)$), as the HOT-Jass Geometry arrows show; B is along $I_p$. In the poloidal $(R, Z)$ view $+\phi$ points into the page and $\mathbf{B}_p = B_p\, \hat{e}_\phi \times \hat{n}$ ($\hat{n}$ the outward normal of the elliptical flux surface) circulates clockwise: outward at the top, down on the outboard side.

Reversed fields: `PlasmaParams.ip_sign` / `bt_sign = -1` (HOT-Jass: negative Ip / B0 inputs) reverse $I_p$ (and $\mathbf{B}_p$ with it) or $B_t$. The plasma current and field stay magnitudes in every scaling, q, orbit-width and Greenwald formula; the signs enter only the birth pitch, and co-current beams

always run along $I_p$. The pitch is returned relative to $I_p$ ($\xi_0 = \sigma_I \sigma_B \xi_{lab}$), so only the relative direction of $B_t$ and $I_p$ matters: reversing $I_p$ alone is the mirror image of the machine in a vertical plane (identical results), reversing both is the machine turned upside down (up- and down-inclined beams, and $\pm Z_t$, swap), and reversing $B_t$ alone is therefore equivalent to swapping the up and down inclinations.

The birth pitch is taken against the full field (`physics.beam_birth_pitch`),

$$\xi_0 = \frac{\mathbf{v}\cdot\mathbf{B}}{vB} = \frac{\cos\alpha\,(y/R)\,B_R \pm \cos\alpha\,(R_t/R)\,B_\phi + \sin\alpha\,B_Z}{\sqrt{B_\phi^2 + B_p^2}}, \qquad B_\phi = B_t R_0/R,$$

$\pm$ for co / counter beams, with $B_p(\rho) = \mu_0 I(\rho)/L_p(\rho)$ from the assumed current profile $j \propto (1-\rho^2)^\nu$, $\nu = q_{cyl}/q_0 - 1$ (`physics.poloidal_field_T`, the same profile as the Current tab; `TokamakConfig.q0`). The $B_R$ and $B_Z$ terms make an up- and a down-inclined beam differ, and the normalisation by $|B|$ lowers $|\xi_0|$ below the toroidal direction cosine $R_t/R$ by $B_\phi/B$. $\xi_0$ is used by the pitch-resolved ST orbit loss (`first_orbit_loss_fraction_st(..., pitch=)`), the consistent birth weighting and NBCD; an ion born with $\xi_0 < 0$ moves against $I_p$ and is treated as counter-current.

HOT-Jass web, Beam tab: the *Fast-ion birth vs normalised radius* panel also plots each NBI's birth-weighted mean pitch $\xi_0(\rho)$ for its full-energy component (dash-dot, right axis), from the same deposition model (`beam_birth_density_profile(..., return_pitch=True)`, with $q_0$ and the $I_p/B_t$ signs the solver uses); it is blanked where the beam deposits less than 2 % of its peak.

### 1.5 Beam energy components (full, half, third)

A positive-ion source also extracts D2+ and D3+ (H2+/H3+, T2+/T3+), which dissociate in the neutraliser into atoms at $E_b/2$ and $E_b/3$. Each beam therefore carries **power fractions** $f_1, f_2, f_3$ of its full-, half- and third-energy components (`BeamParams.energy_fractions`, default $(1, 0, 0)$ – a mono-energetic beam):

$$E_k = E_b/k, \qquad P_k = f_k\,P_{NB}, \qquad k = 1, 2, 3.$$

`HotJassModel` splits every beam into its non-zero components before the solve (`beam_components()`), so shine-through, orbit and CX losses, slowing-down, beam-target fusion and current drive all run per component, and the per-beam result lists (`f_capture`, `I_nb_per_beam_A`, ...) hold one entry per component. The particle fractions follow from $N_k \propto P_k/E_k$, i.e. $\propto k f_k$. Beam-beam fusion sums over every component pair of the two beamlines.

### 1.6 Two-beam slowing-down model

The slowing-down quantities are calculated independently for each beam. For beam $j$:

$$E_{c,j} = \left(\frac{m_j}{m_e}\right)^{1/3} T_e$$

$$\tau_{se,j} = 6.27 \times 10^8\,\frac{A_j T_e[\mathrm{eV}]^{3/2}}{n_e[\mathrm{cm}^{-3}]\ln\Lambda}$$

$$\tau_{S,j} = \frac{\tau_{se,j}}{3}\ln\left[1 + \left(\frac{E_{b,j}}{E_{c,j}}\right)^{3/2}\right]$$

The useful beam power includes the HotJass capture and loss cascade. The fast-ion density from each beam is:

$$n_{b0,j} = \frac{P_{\text{useful},j}\tau_{S,j}}{E_{b,j}(10^3 e)V}$$

The total fast-ion density is:

$$n_{b0} = \sum_j n_{b0,j}$$

## 1.7 Density peaking and shine-through

The density peaking input defines the analytic electron-density profile:

$$n_e(\rho) = n_{e0}(1-\rho^2)^{2p_n}, \qquad 0 \le \rho \le 1$$

Shine-through uses the density along the beam chord rather than assuming that the central density applies everywhere. Each NBI supplies a tangent point $(R_t, Z_t)$ in the NBI tab. The straight tangential chord at that point is intersected with the shaped plasma cross-section, and the optical depth is evaluated as:

$$\tau_b = \sigma_{\text{stop}}(E_b/A_b)\int_{\text{chord}} n_e(\rho(s))\,ds$$

and the captured and shine-through fractions are:

$$f_{\text{capt},b} = 1 - e^{-\tau_b}, \qquad f_{\text{shine},b} = e^{-\tau_b}.$$

The chord is evaluated numerically using the local cylindrical radius along the tangent path and the plasma elongation. With $p_n = 0$, the density is uniform along the computed chord; with $p_n > 0$, the profile is integrated point by point. A tangent point outside the plasma produces zero captured power.

### 1.7.1 Shine-through model selection

Each NBI has a three-way shine-through selector:

- **Riviere**: the existing Riviere-type analytic stopping cross-section fit.
- **Janev/Suzuki**: the Janev-Boley-Post analytic fit from the supplied paper, Nuclear Fusion 29 (1989) 2125, Eqs. (23)-(26), using the published Table III coefficients and the carbon-impurity coefficients from Table IV. It models total effective stopping, including the paper's density, electron-temperature and $Z_{\text{eff}}$ dependence. The fit is published for $100 \le E \le 10^4$ keV/u, $10^{12} \le n_e \le 10^{15}$ cm$^{-3}$ and $1 \le T_e \le 50$ keV; inputs outside these ranges are clipped to the nearest validity boundary.
- **Manual**: the entered manual shine-through fraction is used directly; captured power is $(1 - f_{\text{shine}})P_{NB}$.

The selected calculated stopping model is also used by the optional first-orbit-loss attenuation weighting. The Manual mode bypasses the cross-section calculation for shine-through.

## 1.8 First-orbit loss (reduced model)

First-orbit loss is an **opt-in** channel (the *First-orbit loss* checkbox in the Losses section). When enabled, it estimates the fraction of *captured* fast ions that leave the plasma promptly on their first drift orbit, before any slowing-down or charge-exchange because they are born close enough to the last closed flux surface (LCFS) that their guiding-centre orbit, or their gyro-orbit, crosses it.

The **Orbit model** selector (Losses section) chooses how the orbit widths and the loss criterion are built:

- **Large-aspect ($q_*\rho_{Li}$)** the original reduced model described in *Passing-orbit width* and *Loss criterion* below. Direction-only mean shift; exactly zero co-current loss.
- **ST orbits mean-shift (arbitrary $A$)** and **ST orbits pitch-resolved** two spherical-tokamak variants described in *ST orbit models* at the end of this section. Both give non-zero co-current loss.
- **ST orbits $\mathbf{P}_\phi$ orbit boundary** births followed along their guiding-centre orbits from the constants of motion (*Orbit-boundary model* at the end of this section). In HOT-Jass web it is the default of the spherical presets (SANTE, ST40, MAST-U, NSTX-U); JET and TCV default to the pitch-resolved model, ITER and T-15MD to the large-aspect one.

### 1.8.1 Birth profile along the chord

The calculation reuses the shine-through deposition. Along the straight tangential chord (assumed $R_t = R_0$, length $2a$) the local ionisation rate density is

$$\dot{n}(x) \propto n_{e0}\,\sigma_{\mathrm{stop}}(E_{b,j}/A_j)\,e^{-n_{e0}\sigma_{\mathrm{stop}}x}, \qquad x \in [0, 2a],$$

and the path coordinate is mapped to a normalised flux label by

$$\rho(x) = \frac{|x - a|}{a},$$

so $\rho = 0$ at the chord mid-point and $\rho = 1$ at both ends. The stopping cross-section is the beam's selected shine-through model; **Manual** mode has no deposition shape, so the Riviere fit is substituted for this weighting only.

**Off-axis / vertically-shifted injection.** When a beam's tangency radius $R_t$ differs from $R_0$, or its aim height $Z_t \neq 0$, the birth profile is built instead from the *real* chord: a horizontal line tangent to the cylinder $R = R_t$ at height $Z_t$, sampled through elliptical flux surfaces

$$\rho(s)^2 = \left(\frac{R(s) - R_0}{a}\right)^2 + \left(\frac{Z_t}{\kappa a}\right)^2, \qquad R(s) = \sqrt{R_t^2 + (s - s_0)^2},$$

with the chord length running edge-to-edge and $\rho$ no longer reaching 0 (the beam no longer passes through the magnetic axis). This is the same geometry the shine-through / capture calculation already uses. For the standard on-axis tangential case ($Z_t = 0$, $R_t = R_0$ or unset) the crude $\rho(x) = |x - a|/a$ mapping above is kept, so existing results are unchanged.

**Central column.** The beam enters from the outboard side. If its tangency radius is inside the central column ($R_t < R_{\mathrm{post}}$, set by `Centre-post R`; default $R_0 - a$), the straight chord would strike the column, so only the **outboard leg** from the outboard-edge entry to the point where $R(s) = R_{\mathrm{post}}$ is kept; the (unreachable) far leg is discarded. A beam with $R_t$ at or outboard of

$R_{\mathrm{post}}$ threads the gap between the column and the inboard plasma edge and keeps its full chord. A beam whose aim point lies outside the plasma altogether contributes no first-orbit loss. The Deposition tab's geometry panel draws the column and marks a blocked beam.

### 1.8.2 Passing-orbit width

The radial width of a strongly co-passing drift orbit at the full injection energy is the textbook large-aspect-ratio estimate

$$\Delta r_j = q_*\,\rho_{Li,j}, \qquad q_* = \frac{2\pi a^2 B_t}{\mu_0 R_0 I_p}\,\frac{1+\kappa^2}{2}, \qquad \rho_{Li,j} = \frac{m_j v_{b,j}}{eB_t}.$$

$q_*$ is the cylindrical, elongation-corrected safety factor and $\rho_{Li,j}$ the toroidal-field gyroradius at $E_{b,j}$.

### 1.8.3 Loss criterion

The orbit is modelled as a **rigid mean shift** of the birth radius: inward for co-current injection (favourable), outward for counter-current:

$$\rho_{\mathrm{eff}}(x) = \rho(x) \mp \frac{\Delta r_j}{a} \qquad (\mathrm{co}\,/\,\mathrm{counter}),$$

and an ion born at $x$ is counted as lost if $\rho_{\mathrm{eff}}(x) > 1$. Because $\rho(x) \le 1$, the co-current shift can never trigger this test, so the mean-shift model gives **exactly zero first-orbit loss for co-current injection**; for counter-current it loses every ion born with $\rho(x) > 1 - \Delta r_j/a$.

An optional, direction-independent **Larmor (gyro-orbit) prompt-loss** channel can be OR-ed in: an ion whose guiding centre is born within one gyro-diameter of the LCFS,

$$\rho(x) > 1 - \frac{2\rho_{Li,j}}{a},$$

is lost regardless of the drift direction (gyration is much faster than the guiding-centre drift). This channel is disabled by default.

### 1.8.4 Loss fraction and power

$$f_{\mathrm{orbit},j} = \frac{\int_{\mathrm{lost}\ x} \dot{n}(x)\,dx}{\int_0^{2a} \dot{n}(x)\,dx}, \qquad P_{\mathrm{orbit},j} = f_{\mathrm{orbit},j}\,P_{\mathrm{capt},j}.$$

First-orbit loss is applied to the captured power *before* the charge-exchange loss and before the useful power that drives $T_e$, $T_i$ and $n_{b0,j}$. Injection direction (co-/counter-current) is set per NBI in its input section.

### 1.8.5 Reported quantities

The Dashboard and the Assumptions tab report, per beam, $\rho_{Li,j}$ and $\Delta r_j$ (in cm and as fractions of $a$) and $f_{\mathrm{orbit},j}$.

### 1.8.6 Limitations

- **The co-current zero is a model artefact, not physics.** The rigid mean-shift ignores the finite radial width of the orbit and the outboard excursion that a favourable inward mean shift does not cancel; published orbit-following comparisons give of order a few percent co-current first-orbit loss, versus tens of percent counter-current.
- The chord label $\rho(x) = |x - a|/a$ folds the inboard and outboard halves together, so the model cannot distinguish a high-field-side birth from a low-field-side birth. It therefore cannot represent the correct structure co-current losing high-field-side edge births, counter-current losing low-field-side edge births, and trapped-ion banana-tip losses on the low-field side in either direction.
- Circular / elliptical cross-section, large-aspect-ratio orbit widths (the birth chord itself now honours $R_t$ and $Z_t$, but the orbit-width and pitch estimates are still evaluated at $R_0$); $\Delta r_j$ and $\rho_{Li,j}$ are evaluated at the full injection energy (an upper bound real ions slow down).
- The birth-profile integral in this channel uses the flat central density $n_{e0}$, not the peaked profile used elsewhere.
- When $\rho_{Li}/a$ or $\Delta r/a \gtrsim 0.3$ orbit width comparable to the machine size, typical of compact low-aspect-ratio devices the whole estimate is order-unity uncertain and really needs a guiding-centre orbit follower. The Assumptions tab flags this case. The ST orbit models below address the first three of these limitations directly.

### 1.8.7 ST orbit models (arbitrary aspect ratio)

On a spherical tokamak the drift and banana orbit widths are an $O(a)$ fraction of the minor radius, the trapped fraction is large, and outboard-born trapped ions are lost **regardless of injection direction** so the large-aspect model's exact co-current zero is qualitatively wrong. The two ST variants build their widths from the **poloidal gyroradius**

$$\rho_{\theta,j} = \frac{B_t}{\bar{B}_p}\,\rho_{Li,j}, \qquad \bar{B}_p = \frac{\mu_0 I_p}{2\pi a\sqrt{(1+\kappa^2)/2}},$$

(not $q_*\rho_{Li}$, whose $q_*$ runs away as $\varepsilon \to 1$), together with the circulating and banana widths and the arbitrary-$A$ trapped fraction

$$w_{\mathrm{pass},j} = \varepsilon\,\rho_{\theta,j}, \qquad w_{\mathrm{ban},j} = 2\sqrt{\varepsilon}\,\rho_{\theta,j}, \qquad f_t = 1 - \frac{(1-\varepsilon)^2}{(1+1.46\sqrt{\varepsilon})\sqrt{1-\varepsilon^2}},$$

with $\varepsilon = a/R_0$ (clamped to $[10^{-3}, 0.95]$) and $f_c = 1 - f_t$ (Lin-Liu & Miller, *Phys. Plasmas* **2** (1995) 1666). An arbitrary-$A$ edge safety factor $q_a$ (Uckan / ITER Physics Basis form, with the $(1.17 - 0.65\varepsilon)/(1-\varepsilon^2)^2$ shaping factor) is reported for reference but is **not** used in the widths.

The birth profile in this channel uses the same Beer-Lambert deposition density as the large-aspect model and the Deposition tab's chord plot, $n_e\sigma\, e^{-n_e\sigma x}$ via the selected stopping model so it responds to $n_{e0}$ the same way everywhere in the code (an earlier version used a fixed "penetration rule of thumb" mean free path that saturated at its clamp for realistic densities, decoupling the ST models' loss fraction from $n_{e0}$; this has been fixed).

The **drift-orbit shift is direction-dependent** for both populations inward for co-current ($v_\parallel$ along $I_p$), outward for counter with a weaker shift $c_t\, w_{\mathrm{pass}}$ ($c_t = 0.25$) for the trapped-ion banana centre than for a passing ion. The **gyro channel** ($\rho > 1 - \rho_{Li}/a$, born within one gyroradius of the LCFS) is direction-independent. Co-current loss is therefore non-zero but smaller than counter-current the co/counter asymmetry survives, unlike the large-aspect model's all-or-nothing 0 vs 1.

**Mean-shift variant.** Deposition-weighted loss integrals over $\rho(x)$, blended by $f_c$, $f_t$ (sign convention: − co, + counter):

$$f_{\mathrm{orbit},j} = f_c\, L_{\mathrm{pass}} + f_t\, L_{\mathrm{trap}},$$

$$L_{\mathrm{pass}}:\ \rho \mp \frac{w_{\mathrm{pass},j}}{a} > 1 \ \text{ or } \ \rho > 1 - \frac{\rho_{Li,j}}{a},$$

$$L_{\mathrm{trap}}:\ \rho + \frac{w_{\mathrm{ban},j}}{2a} \mp \frac{0.25\, w_{\mathrm{pass},j}}{a} > 1 \ \text{ or } \ \rho > 1 - \frac{\rho_{Li,j}}{a}.$$

**Pitch-resolved variant.** A 2-D integral over birth radius $x$ and the pitch **magnitude** $|\lambda| = |v_\parallel/v|$. The birth value $|\lambda_0(x)| = R_t/\sqrt{R_t^2 + (x-a)^2}$ is fixed by the injection *geometry* and does **not** depend on the poloidal-field / current direction only the sign of $v_\parallel$ relative to $I_p$ flips (co ↔ counter), carried by $s_{\mathrm{dir}} = \mp 1$. $|\lambda_0|$ is smeared by a $\sigma = 0.10$ Gaussian. An ion is trapped if $|\lambda| < \sqrt{2\varepsilon_{\mathrm{loc}}/(1+\varepsilon_{\mathrm{loc}})}$ ($\varepsilon_{\mathrm{loc}} = \rho\, a/R_0$); its outboard radial excursion is

$$\text{passing}:\ s_{\mathrm{dir}}\, \frac{w_{\mathrm{pass},j}}{a}\, |\lambda|, \qquad \text{trapped}:\ \frac{w_{\mathrm{ban},j}}{2a}\, \frac{|\lambda|}{\lambda_{\mathrm{tr}}} + s_{\mathrm{dir}}\, \frac{0.25\, w_{\mathrm{pass},j}}{a},$$

(the banana half-width → 0 for deeply trapped ions, largest near the trapped/passing boundary), and it is lost if $\rho$ + (excursion) $> 1$ or $\rho > 1 - \rho_{Li,j}/a$.

Both variants collapse toward the large-aspect co-current zero as $\varepsilon \to 0$. They remain 0-D order-of-magnitude estimates absolute magnitudes are upper-bound-like and, because the direction-independent trapped banana-tip and gyro channels stay significant, the co/counter split is narrower than the large-aspect model's (typically a factor $\sim 1.5$, not 0 vs 1). References: Akers *et al.*, *Nucl. Fusion* (START NBI, $A \sim 1.4$); Goldston, White & Boozer, *Phys. Rev. Lett.* **47** (1981) 1004; Goldston & Rutherford, *Introduction to Plasma Physics* (1995), Ch. 12.

### 1.8.8 Orbit-boundary model from the constants of motion (`orbit_model = "st_pphi"`)

HOT-Jass: MODELS, "ST orbits - $P_\phi$ orbit boundary". Instead of estimating drift and banana widths, each birth is followed along its guiding-centre orbit (`physics.pphi_orbit_loss_probability`). Energy, magnetic moment $\mu = mv_\perp^2/2B$ and canonical toroidal angular momentum

$$P_\phi = mRv_\phi - e\Psi$$

are conserved, with $\Psi(\rho)$ the poloidal flux per radian, increasing outward relative to $I_p$, $d\Psi/d\rho = R_0 B_p(\rho)\, a\sqrt{(1+\kappa^2)/2}$ (`physics.poloidal_flux_profile`, $B_p$ from the assumed current profile, see "Field directions and the beam birth pitch"). With $B \approx B_0 R_0/R$ and $v_\phi \approx v_\parallel$, a birth at $(R_b, \Psi_b)$ with signed pitch $\xi_0$ (positive along $I_p$) sits at major radius $R$ of an orbit leg on

$$\Psi(R) = \Psi_b + \frac{m}{e}\left[R\, v_\parallel(R) - R_b v\, \xi_0\right], \qquad v_\parallel(R) = \pm v\sqrt{1 - (1-\xi_0^2)\, R_b/R},$$

with the bounce point at $R = R_b(1-\xi_0^2)$. The orbit exists at $R$ where $\Psi(R) \geq \Psi(\rho = |R - R_0|/a)$ (the flux surface reaches that $R$). The birth leg is followed over the connected range of $R$ around $R_b$ where it exists; if that range reaches the bounce point (a trapped ion), the return leg (opposite sign of $v_\parallel$) is added. The ion is lost if the outermost flux reached corresponds to $\rho \geq 1 - \rho_{Li}/a$ (the gyro criterion of the other ST models). There are no fitted constants: a co-current ion born on the outboard side is at the outermost point of its orbit (its banana or drift orbit lies inward)

and is lost only within a Larmor radius of the edge; a counter-current ion is at the innermost point, its orbit reaching a full banana or drift width outward. The birth pitch carries the same 0.1 Gaussian spread (beam divergence, finite chord) as the pitch-resolved model, as a 3-point Gauss-Hermite average.

Implementation: in this up-down symmetric approximation the outward excursion $\rho_{out} - \rho_b$ depends on $(R_b, \rho_b, \xi_0)$ only; it is tabulated once per beam energy and machine on a $33 \times 33 \times 65$ grid (`_pphi_excursion_table`, cached, ˜0.2 s) and interpolated, and the loss criterion is applied per birth, which keeps the loss boundary sharp (table against direct orbits: mean loss within ˜2 %). The lost power fraction is the loss probability averaged over the births along the beam-axis chord, weighted by the birth rate $n_e(\rho)\sigma_s e^{-\tau}$ with the real density profile; the birth profile is always weighted by the same probability (no cutoff radius, the "consistent" option is implied).

Approximations: up-down symmetric elliptical flux surfaces; $|B| \approx B_\phi$ in the bounce condition and $v_\phi \approx v_\parallel$; flux-surface averaged $B_p$ ($|\nabla\psi| \approx R_0 B_p$) from the assumed current profile; no wall or centre-post shape beyond the LCFS; collisions during the first orbit neglected.

Comparison with the pitch-resolved model (presets, beam-axis NBI-1 / NBI-2): co-current losses fall strongly (SANTE 23 % → 9 %, MAST-U 30 % → 15 %, NSTX-U 29 % → 19 %, TCV 24 % → 5 %, JET 4.7 % → 2.4 %, T-15MD 5.4 % → 1.4 %), counter-current ones stay large (SANTE NBI-2 55 %, TCV NBI-2 28 %).

References: Rome & Peng, *Nucl. Fusion* **19** (1979) 1193 (orbit topology and loss boundaries from the constants of motion); Eriksson & Porcelli, *Plasma Phys. Control. Fusion* **43** (2001) R145 (review of energetic-ion orbits); Heidbrink & Sadler, *Nucl. Fusion* **34** (1994) 535 (fast ions in tokamak experiments, prompt losses); White, *The Theory of Toroidally Confined Plasmas* (guiding-centre motion, canonical momentum).

## 1.9 Charge-exchange loss

A captured fast ion can charge-exchange with a background cold neutral during slowing-down, becoming a fast **neutral** that escapes the plasma before fully thermalizing a second loss channel, applied to the post-orbit-loss captured power (Step 0, same ordering as first-orbit loss above). Design note: `docs/CX model.pdf`. A **CX-loss model** selector (rail, "Losses" section) picks between three modes; only the active mode's inputs matter.

### 1.9.1 Manual fraction (default)

The original flat efficiency knob `cx_loss_fraction` $\in [0, 1]$: a fixed fraction of captured power is removed, independent of $E_b$, $n_e$, or any background-neutral density. PPPL-1280 Sect. 6.2.2/6.2.3 quotes "up to 10%" of injected power for $Z_{\text{eff}} = 1$ as a rough qualitative figure, not a formula. Default 0 =off; leaves every other result byte-identical.

### 1.9.2 Manual n0/ne the physics-based survival integral

The user supplies $n_0/n_e$ directly (a spatially uniform, 0-D background neutral density $n_0 = (n_0/n_e)\, n_{e0}$), and the model evaluates the actual CX survival integral along the slowing-down path. For a fast ion at energy $E$:

$$\nu_{\text{cx}}(E) = n_0\, \sigma_{\text{cx}}(E)\, v(E), \qquad S(E) = \exp\left[ - \int_E^{E_b} \frac{\nu_{\text{cx}}(E')}{|dE'/dt|}\, dE' \right],$$

with particle- and power-loss fractions

$$f_{\mathrm{cx},N} = 1 - S(0^+), \qquad f_{\mathrm{cx},P} = \int_0^{E_b} \frac{E}{E_b}\,\frac{\nu_{\mathrm{cx}}(E)}{|dE/dt|}\,S(E)\,dE.$$

$f_{\mathrm{cx},P}$ (the quantity actually applied to $P_{\mathrm{useful}}$) is smaller than $f_{\mathrm{cx},N}$ because $\sigma_{\mathrm{cx}}$ falls steeply above $\sim$30 to 50 keV/amu, so most CX events happen late in the cascade, when little energy is left to remove. $1/|dE/dt| = (\tau_{se}/2)\sqrt{E}/(E^{3/2} + E_c^{3/2})$ is exactly the same slowing-down kernel already used by `thermalization_time/slowing_down_distribution` (differentiating $\tau_s(E) = (\tau_{se}/3)\ln[1 + (E/E_c)^{3/2}]$ reproduces it), so no new slowing-down physics is introduced only the extra $\nu_{\mathrm{cx}}(E)$ factor under the same integral. $\sigma_{\mathrm{cx}}(E)$ is the Janev & Smith (1993, Sect. 2.3.1) analytic form, refit by Swaczyna, Bzowski & Kubiak (arXiv:2411.13174, 2024, their Eq. A1) against Schultz *et al.* (2023) theory and the Barnett (1990) tabulation quoted accuracy 5% from 1100 keV/amu, the range this project's beams live in.

Also reported, per beam, on the Dashboard: the dimensionless figure of merit $\gamma_{\mathrm{cx}} \equiv \nu_{\mathrm{cx}}(E_b)\,\tau_s \approx n_0\sigma_{\mathrm{cx}}(E_b)v_b\tau_s$ negligible for $\gamma_{\mathrm{cx}} \ll 1$, worth taking seriously for $\gamma_{\mathrm{cx}} \sim 0.1$ to 1, and a sign the single-pass loss treatment is breaking down for $\gamma_{\mathrm{cx}} \gtrsim 1$ (flagged with a warning, as is $f_{\mathrm{cx},P} > 10\%$, PPPL-1280's rough $Z_{\mathrm{eff}} = 1$ ceiling). Because CX loss depends on $T_e$ (via $E_c(T_e)$) while $T_e$ itself depends on the CX-reduced $P_{\mathrm{useful}}$, this is a genuine fixed point; an outer under-relaxed Picard loop (in `hotjass_core`, mirroring the existing alpha-heating fixed point) closes it, keeping the core solver a single non-iterative solve.

### 1.9.3 Penetration (n0_LCFS/ne) edge-penetration estimate

Instead of a single uniform $n_0$, the boundary value $n_{0,\mathrm{LCFS}} = (n_{0,\mathrm{LCFS}}/n_e)\,n_{e,\mathrm{LCFS}}$ is set from the user ratio, and the core profile follows the standard penetration/recycling picture. $n_{e,\mathrm{LCFS}}$ is evaluated at $\rho = 0.95$, not literally at $\rho = 1$: the core density shape $n_e(\rho) = n_{e0}(1 - \rho^2)^{2p_n}$ is an exact zero at $\rho = 1$ whenever $p_n > 0$, which would make $n_{0,\mathrm{LCFS}} = 0$ regardless of the input ratio – a knob that does nothing. $\rho = 0.95$ is a compromise: with a strongly peaked profile ($p_n \gtrsim 1$ to 2) it can still sit orders of magnitude below $n_{e0}$, so treat this mode as unreliable for very peaked density profiles.

$$\lambda_0(\rho) = \frac{v_0}{n_e(\rho)\,[\langle\sigma v\rangle_{\mathrm{ion}}(T_e) + \langle\sigma v\rangle_{\mathrm{cx}}(T_i)]}, \qquad n_0(\rho) \approx n_{0,\mathrm{LCFS}}\exp\left[-\int_\rho^1 \frac{a}{\lambda_0(\rho')}\,d\rho'\right],$$

volume-averaged to a single $\langle n_0\rangle$ and fed through the same survival integral as "Manual n0/ne" above. $v_0$ is a round $\sim 3$ eV Franck-Condon-like dissociation speed. **Caveat:** the ionization ($\langle\sigma v\rangle_{\mathrm{ion}}(T_e)$) and thermal-CX ($\langle\sigma v\rangle_{\mathrm{cx}}(T_i)$) rate coefficients used for $\lambda_0$ are an interim hand-built approximation (a threshold-shaped form for ionization, a threshold-free $\sqrt{T}$ form for CX, each capped to saturate at a few $\times 10^{-14}\,\mathrm{m}^3/\mathrm{s}$) a calibrated Voronov (1997) / ADAS `adf11` fit was not sourced/verified in this session, so treat $n_0(\rho)$'s absolute scale as order-of-magnitude only; the survival-integral physics downstream of $\langle n_0\rangle$ (same $\sigma_{\mathrm{cx}}(E)$ fit as "Manual n0/ne") is not affected by this caveat. $n_0(\rho)$ itself (per the active mode – flat for "manual n0/ne", the profile above for "penetration", nothing for "manual fraction") is plotted on the Profiles tab; because $\lambda_0$ is typically much smaller than $a$, the profile can span many orders of magnitude edge-to-core (shown on a log axis) – a genuine, if extreme, feature of this reduced model, not a numerical artifact.

### 1.9.4 Escape probability (informational only)

A fast neutral born from a CX event can re-ionize before reaching the wall, so treating every CX event as a full loss is a conservative upper bound, not exact. An escape-probability factor

$\exp(-x_{\rm wall}/\lambda_{\rm reion})$ is shown per beam (Dashboard) $x_{\rm wall} \approx a$, $\lambda_{\rm reion}$ from the same combined ionization+CX stopping cross section a beam neutral itself uses but is **not** applied to $f_{{\rm cx},P}$; it is informational, per the design note's explicit "acceptable as the conservative default, but shown as such."

## 1.10 Bulk toroidal rotation

A **Rotation model** selector (Losses section, below the CX-loss fields) picks between three modes, off by default – like every other loss/model knob, this leaves every existing result byte-identical unless explicitly enabled.

**Where rotation matters and where it doesn't.** $v_\phi$ (typically $\sim 10^2$ to $10^3$ km/s) is small next to the beam's own velocity ($\sim 3$ to $5 \times 10^3$ km/s for a 100180 keV D/T beam), so shine-through, first-orbit loss and CX loss all governed by the beam's own velocity are essentially insensitive to it and are **not** touched. The one channel where it matters cleanly is **beam-target fusion**: that reaction's cross-section and flux factor depend on the beam-target *relative* velocity, which the un-rotated model implicitly assumes equals the beam's lab velocity (a stationary target). With rotation:

$$v_{\rm rel}(E) = |\, v_{\rm beam}(E) - s\, v_\phi \,|\,, \qquad s = +1 \text{ (co-current beam)},\ -1 \text{ (counter)},$$

used for *both* the flux factor and the cross-section's energy argument (re-expressed as the equivalent deuteron energy $E_D = \frac{1}{2} M_D v_{\rm rel}^2$, the same conversion `beam_target_reactivity_spectrum()` already used for a T beam). $v_\phi = 0$ reproduces the original stationary-target result exactly.

**Non-monotonic, not just a small correction in one direction.** The D-T cross-section peaks around $\sim 64$ keV (deuteron-equivalent energy) and falls off on both sides. For a beam energy *above* that peak (this project's 120180 keV presets all are), pulling $v_{\rm rel}$ down (co-current beam, co-rotating plasma) moves the *high-energy tail* of the slowing-down spectrum back toward the peak and can *increase* beam-target yield, while a counter-current beam sees the opposite – the sign is not "co always suppresses fusion", it depends on where $E_b$ sits relative to the cross-section peak, and the net effect is an integral over the whole slowing-down spectrum (not just $E_b$ itself), evaluated numerically.

### 1.10.1 Manual $v_\phi$

Direct input of a single, spatially-uniform $v_\phi$ (m/s, positive = co-current) – the honest option when the collisionality of the source is unknown but a plausible rotation speed is.

### 1.10.2 Momentum balance

$v_\phi$ instead comes from a 0-D toroidal angular-momentum balance, the rotation analogue of this project's energy balance:

$$T_{\rm NBI} = \sum_j s_j \, \frac{2 P_{{\rm useful},j}\, R_{t,j}}{v_{b,j}}, \qquad v_\phi = \frac{T_{\rm NBI}\, \tau_\phi}{\rho_i\, R_0\, V},$$

summing each beam's injected canonical angular momentum ($s_j = +1$ co, $-1$ counter; $R_{t,j}$ its own tangency radius), balanced against a momentum-confinement loss $L/\tau_\phi$ with $L \sim \rho_i v_\phi R_0 V$ ($\rho_i$ = thermal-ion mass density, $n_{D0} M_D + n_{T0} M_T$) the bulk plasma treated as a single representative rotating mass at the magnetic axis, the same level of 0-D reduction as everywhere else

in this model. **Caveat:** unlike $\tau_E$, there is no widely validated scaling for the momentum confinement time $\tau_\phi$; the user-set ratio `tau_phi / tauE,i` (default 1.0, the common crude $\tau_\phi \approx \tau_{E,i}$ assumption) sets $\tau_\phi = (\text{ratio}) \times \tau_{E,i}$, so treat $v_\phi$'s absolute scale here as order-of-magnitude, more so than the manual-input mode.

The Dashboard reports $v_\phi$ (km/s) and a Mach number $M = v_\phi / c_s$ ($c_s = \sqrt{(T_{e0} + T_{i0})/m_i}$, a simple ion-acoustic estimate, not a rigorous sound speed), plus the net torque $T_{\mathrm{NBI}}$ for the momentum-balance mode.

### 1.11 Beam-beam fusion

Off by default (checkbox, Losses section, below the rotation fields). Fast ions from **two separate NBI sources** can fuse with each other directly – a third fusion channel alongside thermal-thermal and beam-target. This is a **reduced** model, not a full treatment: instead of integrating over both beams' slowing-down spectra (a genuine 2-D relative-velocity integral), each population is approximated by a single characteristic velocity at its own average fast-ion energy (`average_fast_energy_keV`, the same representative energy this project already uses for pressure/energy-density diagnostics), signed by its own co/counter direction. Only the **first two** beamlines are paired – this project's tested scope is exactly two NBI sources – summed over every pair of their energy components.

The relevant physical quantity is the beam-beam **relative** velocity:

$$v_{\mathrm{rel}} = |\, s_1 v_1 - s_2 v_2 \,|, \qquad s = +1 \text{ (co)},\ -1 \text{ (counter)},$$

fed into the same deuteron-equivalent-energy cross-section evaluation as the rotation correction to beam-target fusion (Sec. "Bulk toroidal rotation" above) – structurally the same construction, with the *other beam's own velocity* playing the role $v_\phi$ played there. **This is why counter-injecting the second beam matters physically**: two co-current beams at similar per-amu energy have a small $v_{\mathrm{rel}}$ (their velocities nearly cancel) and thus a negligible beam-beam yield, while opposite directions maximize $v_{\mathrm{rel}}$ – exactly DANTE's default configuration (D co-current, T counter-current). D-T and D-D pairs are handled (the only cross-section fits this project carries); T-T pairs, or a beam-beam calculation with fewer than two useful beams, contribute nothing.

The result is folded into the existing D-T/D-D fusion power and neutron- rate totals (`pf_dt_w`, `pf_dd_w`, `neutron_rate_s`) exactly like the thermal and beam-target contributions, so it is automatically included in the alpha-heating closure (Sec. "Alpha self-heating" below) with no separate wiring needed. The Dashboard's D-T/D-D fusion-power rows show all three contributions (thermal / beam-target / beam-beam); the Scan tab's "Neutrons" group breaks the neutron rate out the same way, with the beam-beam panel showing a genuine (if usually small, for DANTE's default 0.1 MW counter-beam) nonzero curve once enabled, rather than the flat zero it shows while off.

### 1.12 Power balance (electron and ion)

At each central density the model solves a 0-D steady-state power balance for $T_e$ and $T_i$. The heating that drives each temperature is the sum of the neutral-beam split, prescribed auxiliary heating (ECRH and ICRH), and when enabled fusion alpha self-heating.

### 1.12.1 Beam heating split

Each useful beam $j$ delivers a fraction $L_e(x_j)$ of its power to electrons and $L_i(x_j) = 1 - L_e(x_j)$ to ions, with $x_j = E_{b,j}/E_{c,j}(T_e)$ and

$$L_i(x) = \frac{1}{x}\int_0^x \frac{du}{1+u^{3/2}}, \qquad L_e(x) = 1 - L_i(x).$$

$L_i \to 1$ as $x \to 0$ (cold, ion-dominated) and $L_i \to 0$ as $x \to \infty$ (fast beam, electron-dominated). $P_{\text{useful},j}$ is the injected power after shine-through, first-orbit loss and charge-exchange loss (preceding sections).

### 1.12.2 Auxiliary heating (ECRH, ICRH)

ECRH and ICRH are entered as powers with electron/ion fractions. Their contribution to each channel is

$$P_{\text{aux},e} = f_e^{\text{EC}} P_{\text{ECRH}} + f_e^{\text{IC}} P_{\text{ICRH}}, \qquad P_{\text{aux},i} = \left(1 - f_e^{\text{EC}}\right) P_{\text{ECRH}} + f_i^{\text{IC}} P_{\text{ICRH}}.$$

ECRH is electron-dominant, $f_e^{\text{EC}}$ close to 1. For ICRH the electron and ion fractions are set independently; if $f_e^{\text{IC}} + f_i^{\text{IC}} < 1$ the remainder is treated as power that does not thermalise (a fast-ion tail or direct loss). These sources are Te/Ti-independent constants added to the right-hand side of each balance.

### 1.12.3 Alpha self-heating (optional)

When enabled, 3.5 MeV fusion alphas deposit

$$P_\alpha = f_\alpha \frac{E_\alpha}{E_f} P_{f,\text{tot}} = f_\alpha \frac{3.5}{17.6} P_{f,\text{tot}},$$

split between electrons and ions by the alpha slowing-down fraction

$$L_e^\alpha = L_e\!\left(\frac{E_\alpha}{E_c^\alpha(T_e)}\right), \qquad E_c^\alpha = \left(\frac{m_\alpha}{m_e}\right)^{1/3} T_e,$$

so $P_{\alpha,e} = L_e^\alpha P_\alpha$ and $P_{\alpha,i} = (1 - L_e^\alpha) P_\alpha$. Because $P_\alpha$ depends on $P_{f,\text{tot}}$, which depends on $T_i$, which depends on $P_{\alpha,i}$, the alpha term is closed by an under-relaxed fixed-point iteration; with alpha heating off the solve is a single pass.

### 1.12.4 Decoupled balance (default)

With electron-ion equipartition off, the two balances are solved in sequence. The electron balance is a 1-D root-find for $T_e$:

$$\frac{3}{2} n_{e0} T_e \, (10^3 e) \, \frac{V}{\tau_{E,e}} = \sum_j L_e(x_j) \, P_{\text{useful},j} + P_{\text{aux},e} + P_{\alpha,e}.$$

The ion temperature then follows in closed form (or, for the neoclassical-ion option, from a second 1-D root-find see *Confinement time*):

$$\frac{3}{2}\, n_{\text{thermal}}\, T_i\, (10^3 e)\, \frac{V}{\tau_{E,i}} = \sum_j L_i(x_j)\, P_{\text{useful},j} + P_{\text{aux},i} + P_{\alpha,i},$$

where $n_{\text{thermal}} = n_{\text{sum}}(n_{e0}, Z_{\text{eff}}) - n_{b0}$ is the thermal-ion density left after the fast ions are subtracted. A non-positive $n_{\text{thermal}}$ marks the operating point infeasible.

### 1.12.5 Coupled balance (equipartition on)

When electron-ion equipartition is enabled, the exchange power $P_{ei}(T_e, T_i)$ (defined in *Electron-ion exchange time*) is subtracted from the electron balance and added to the ion balance, and the pair is solved together:

$$\frac{3}{2} n_{e0} T_e (10^3 e) \frac{V}{\tau_{E,e}} = \sum_j L_e(x_j) P_{\text{useful},j} + P_{\text{aux},e} + P_{\alpha,e} - P_{ei}(T_e, T_i),$$

$$\frac{3}{2} n_{\text{thermal}} T_i (10^3 e) \frac{V}{\tau_{E,i}} = \sum_j L_i(x_j) P_{\text{useful},j} + P_{\text{aux},i} + P_{\alpha,i} + P_{ei}(T_e, T_i).$$

The reported $P_e$ and $P_i$ remain the pure beam split; $P_{ei}$, $P_\alpha$ and $P_{\text{aux}}$ are reported separately.

### 1.12.6 Profile-corrected 0-D

The balance above is genuinely 0-D: one density, one $T_e$, one $T_i$, treated as uniform over the volume. With the **profile-corrected 0-D** checkbox (Plasma section) *off*, the entered central density is used directly as that uniform value and the solved $T$ is a flat-plasma effective temperature; comparing it to a measured on-axis $T_0$ then needs a peaking factor supplied by the reader (validation runs use $\simeq 2$). In this mode every radial integral (shine-through column density, thermal and beam-target fusion, pressure, thermal energy) also uses **flat** profiles, consistent with the balance: the peaking parameters are ignored. (Before October 2026 these integrals still applied the $(1-\rho^2)^{2p}$ shapes and treated the flat values as on-axis peaks, which with $p_{Te} = 1$ cut the profile-average $T$ to a third and under-reported fusion power by up to $\sim 3\times$.)

With the checkbox *on*, the entered `central n_e` is taken to be the **on-axis** $n_{e0}$, and for a $(1-\rho^2)^{2p}$ profile the volume average is $\langle X \rangle = X_0/(1+2p)$. The energy balance and the $\tau_E$ scalings then run on

$$\langle n_e \rangle = \frac{n_{e0}}{1 + 2\, p_n},$$

so the solved $T_e$, $T_i$ are **volume-averaged**. The electron and ion temperature profiles are shaped by **independent** exponents $p_{Te}$ (`Temp peaking (electron)`) and $p_{Ti}$ (`Temp peaking (ion)`; a value $< 0$ means "use $p_{Te}$"), so the on-axis values are

$$T_{e0} = \langle T_e \rangle\, (1 + 2\, p_{Te}), \qquad T_{i0} = \langle T_i \rangle\, (1 + 2\, p_{Ti}).$$

They are reported (Dashboard row "on-axis $T_{e0}/T_{i0}$") and are what the fusion, pressure and $\beta_t$ integrals use as their central values  the reactivity integrals with the $(1-\rho^2)^{2p_{Ti}}$ ion shape, the beam slowing-down kernel with the $(1-\rho^2)^{2p_{Te}}$ electron shape. Set $p_n$ (`Density peaking`), $p_{Te}$ and $p_{Ti}$ to match the real profiles; $p \simeq 0.5$ to 0.75 gives $X_0/\langle X \rangle \simeq 2$ to 2.5 (L-/H-mode), while a **hot-ion mode** on a compact ST needs $p_{Ti} \gg p_{Te}$ ($T_{i0}/\langle T_i \rangle \simeq 3$ to 4 with $T_e$ nearly flat). With all peaking parameters at 0 the correction is a no-op.

### 1.12.7 Deposition-weighted fast-ion profile

With profile-corrected 0-D on, the fast-ion density is built **locally** from the beam deposition (October 2026) instead of a single 0-D value spread like the slowing-down time:

$$n_b(\rho) = \sum_j S_j \, h_j(\rho) \, \tau_{s,j}\big(n_e(\rho), T_e(\rho)\big), \qquad S_j = \frac{P_{\mathrm{useful},j}}{E_{b,j}\, e},$$

with $\tau_{s,j}$ the local thermalisation time and $h_j(\rho)$ the confined birth rate per unit volume of beam $j$, normalised so that $V\langle h_j\rangle = 1$ (`physics.beam_birth_density_profile`):

- **Attenuation** along tangential chords (Beer-Lambert) on the $n_{e0}(1-\rho^2)^{2p_n}$ profile, with the same stopping cross-section as the shine-through model.
- **Beam cross-section** (`BeamParams.beam_diameter_m`, HOT-Jass: NBI "diameter"). $D = 0$ (default) is a single pencil ray. $D > 0$ is a circular beam with a Gaussian power profile $p(r) \propto \exp[-(2r/D)^2]$ ($D$ the 1/e diameter, $\sigma = D/2\sqrt{2}$), sampled by parallel rays shifted in tangency radius and height (`physics.beam_ray_bundle`): a square grid over $\pm 3\sigma$, clipped to the $3\sigma$ circle, with Gaussian weights and spacing $\leq 0.1\,a$ (odd, 3 to 21 rays per axis). A single tangential ray puts a finite number of ions into the vanishing volume of its tangency surface $\rho(R_t)$, a singular birth density there; the rays have to be closer than the profile smoothing scale (orbit width, $\geq 0.03$) for their singular profiles to add up to a smooth one. Against a 6500-ray reference the grid is within 1.5 % (L1) and ˜10 % on axis for $D$ up to $0.8a$. All rays are evaluated in one vectorised pass (`physics.tangential_chords`), a few ms per beam: a $D = 0.2$ m beam on a 0.55 m plasma uses 49 rays in 3 ms, against 7 ms for the previous fixed $5 \times 5$ bundle. The shine-through fraction is averaged over the same rays (`captured_power_ fraction(..., beam_diameter_m)`; rays that miss the plasma shine through). First-orbit loss keeps the beam-axis chord.
- **Prompt loss** (already counted in the orbit-loss power), two options (`orbit_loss_deposition`; HOT-Jass: MODELS, "Orbit loss consistent in deposition"):
- `"cutoff"` (default, historical): every birth beyond `orbit_cutoff_rho` is removed. That radius is set by the worst-case (trapped) orbit, so at low current, where the banana width approaches the minor radius, it removes far more births than the loss formula removes power (TCV at 150 kA: 72 % of births against 13 % of power), and the confined beam is renormalised onto the remaining core births.
- `"consistent"` (ST orbit models): each birth is weighted by $1 - P_{lost}(\rho, |\lambda_0|)$, the per-birth rule of the pitch-resolved loss formula (`physics.st_orbit_loss_probability`, also used for the lost power), with $|\lambda_0| = |\xi_0|$ (the full-field birth pitch, "Field directions and the beam birth pitch") along each chord. It changes NBCD by about 30 % on TCV, about 6 % on MAST and 6-19 % on NSTX-U, and is therefore an option, off by default.
- **Orbit-width smoothing.** Births are spread over half the passing orbit width (at least 0.03 in $\rho$), as a *volume-weighted* average of the birth density, so that births do not pile up on the axis.

The volume average of $n_b(\rho)$ replaces the 0-D $n_{b0}$ inside the balance, so dilution, the thermal density $n_{th}(\rho) = n_\Sigma(\rho) - n_b(\rho)$ (locally depleted where the fast ions sit) and the thermal and beam-target fusion integrals all use the same fast ions. The solver stores the radial profiles on `OperatingPoint` (`rho_profile`, `nb_profile_m3`, `n_fuel_profile_m3`, `ne_profile_m3`, `Te_profile_keV`, `Ti_profile_keV`, the fusion power densities, and per beam the birth rates $S_j h_j(\rho)$ and birth pitch $\xi_{0,j}(\rho)$). Their volume integrals are exactly the reported totals. With profile-corrected 0-D off the flat model is kept unchanged.

On the ASCOT5/BBNBI DANTE reference case (4 MW D at 80 keV) the deposition model gives 1.24 MW of fusion power against ASCOT's 1.4 MW; the earlier single-$n_{b0}$ model gave 0.87 MW.

## 1.13 Particle balance and fuelling

The power balance fixes $T_e$ and $T_i$ at a given density, but says nothing about how that density is sustained. The HOT-Jass web app (**Balance** tab) adds a steady-state **particle balance** for the thermal fuel ions as a post-processing step: it uses the solved operating point and does not feed back into the solve.

### 1.13.1 Fast-ion dilution

At fixed $n_e$, quasi-neutrality makes every fast beam ion displace a thermal ion ($n_{\mathrm{thermal}} = n_{\mathrm{sum}} - n_{b0}$, see *Power balance*). The share of fast ions,

$$f_{\mathrm{dil}} = \frac{n_{b0}}{n_{\mathrm{thermal}} + n_{b0}},$$

reduces both thermal and beam-target fusion, which scale with the thermal fuel densities. It is largest for low-energy beams into a moderate-density, hot plasma (long slowing-down time), e.g. $\sim 14\%$ for the DANTE design point.

### 1.13.2 Steady-state balance per species

For each thermal fuel species $s = \mathrm{D}, \mathrm{T}$, holding the input D:T mix at the solved density,

$$S_{\mathrm{ext},s} + S_{\mathrm{beam}\to s} - R_{\mathrm{burn},s} = \frac{N_s}{\tau_p^*}, \qquad N_s = \langle n_s \rangle V, \qquad \tau_p^* = \left(\frac{\tau_p^*}{\tau_E}\right)\tau_{E,e},$$

where

- $N_s$ is the thermal inventory of species $s$ (volume-average density from the balance times the plasma volume), and $N_s/\tau_p^*$ its particle loss rate;
- $\tau_p^*$ is the **effective** particle confinement time including recycling, entered as the ratio $\tau_p^*/\tau_E$ (Plasma input, default 2). In a low-recycling regime $\tau_p^* \approx \tau_p$; with strong recycling $\tau_p^* \gg \tau_p$;
- $S_{\mathrm{beam}\to s}$ is the rate at which beam ions thermalise into their own species (a D beam feeds D): $S_{\mathrm{beam}} = \sum_j P_{\mathrm{useful},j}/(E_{b,j}\, e)$, minus the fast ions burnt in beam-target reactions before they thermalise;
- $R_{\mathrm{burn},s}$ is the thermal ions consumed by fusion: one D and one T per D-T reaction (thermal-thermal reactions consume one of each thermal species; beam-target reactions consume the fast ion and one thermal ion of the other species). D-D burn-up ($\sim 10^{-3}$ of D-T) is neglected;
- $S_{\mathrm{ext},s}$ is the external fuelling (gas puff or pellets) needed to hold the mix. It is also given as a gas throughput, using 1 Pa m$^3$ of $\mathrm{D_2}/\mathrm{T_2}$ at 273 K $= 5.3 \times 10^{20}$ atoms.

### 1.13.3 Over-supplied species

$S_{\mathrm{ext},s} < 0$ means the beam alone delivers more of species $s$ than the plasma loses: the mix can then only be held by pumping. With no external fuelling of that species its thermal fraction settles at

$$f_s = \frac{(S_{\mathrm{beam}\to s} - R_{\mathrm{burn},s})\,\tau_p^*}{N_D + N_T}.$$

This is the typical situation for D beams into a tritium-rich target (DANTE: $S_{\mathrm{beam}}(\mathrm{D}) = 4.85 \times 10^{20}$ s$^{-1}$ against a D loss of $2.85 \times 10^{20}$ s$^{-1}$ at D:T = 10:90, so the D fraction would rise to $\simeq 0.17$

with tritium-only fuelling). Hydrogen beams are reported separately: H is not part of the D:T mix and has to be pumped.

### 1.13.4 Power balance view

The same tab shows the power balance of *Power balance (electron and ion)* as two pies: heating sources ($P_{\mathrm{NB},e}$, $P_{\mathrm{NB},i}$, $P_{\mathrm{aux},e}$, $P_{\mathrm{aux},i}$, $P_\alpha$) and transport losses $W_e/\tau_{E,e}$, $W_i/\tau_{E,i}$, with $W_e = \frac{3}{2}\langle n_e\rangle\langle T_e\rangle V$ and $W_i = \frac{3}{2} n_{\mathrm{thermal}}\langle T_i\rangle V$. In steady state the two losses sum to the total heating; $P_{ei}$ is an internal exchange, not a loss.

### 1.13.5 Limitations

- The result scales directly with $\tau_p^*$, which is an input, not computed; the absolute fuelling rates are only as good as that choice. The sign of $S_{\mathrm{ext},s}$ (whether the beam over-supplies a species) is far less sensitive.
- D and T are assumed to share one $\tau_p^*$; isotope effects on particle transport and wall recycling are not modelled.
- Volume-averaged (0-D) balance: no fuelling-depth or density-profile effects (pellet versus gas puff).

## 1.14 Non-inductive current (NBCD and bootstrap)

The HOT-Jass web app (**Current** tab) estimates the neutral-beam driven and the bootstrap current of a solved operating point (`hotjass.current`, `HotJassModel.current_drive(op)`). It is a **diagnostic only**: nothing feeds back into the solve, so an operating point whose non-inductive current exceeds $I_p$ is flagged rather than corrected. It needs profile-corrected 0-D, because it uses the solver's radial profiles, the beam deposition and the birth pitch; with flat profiles there are no gradients and no current is computed.

### 1.14.1 Neutral-beam current drive

The fast-ion current of each beam is the local 1-D Cordey / Start-Cordey result for a mono-energetic source, with electron and ion drag and ion pitch-angle scattering (no energy diffusion):

$$j_f = Z_b\, e\, S\, h(\rho)\, \tau_s\, v_b\, \xi_0\, J(u_c, \beta), \qquad J = \int_0^1 \frac{u^3}{u^3 + u_c^3} \left[\frac{u^3(1+u_c^3)}{u^3+u_c^3}\right]^{\beta/3} du,$$

where:

- $S\,h(\rho)$ is the confined birth rate per volume (above), and $\tau_s$ the Spitzer slowing-down time ($dv/dt = -(v/\tau_s)(1 + v_c^3/v^3)$).
- $\xi_0$ is the full-field birth pitch ("Field directions and the beam birth pitch"; $\approx R_t/R$ for a horizontal midplane chord with weak $B_p$); it is averaged over the same sub-chords as $h$ and weighted by births, and is negative for counter-current beams.
- $u = v/v_b$, $u_c^3 = (E_c/E_b)^{3/2}$, with the critical energy from the local composition (D, T and the carbon-like impurity), $E_c = 14.8\, A_b T_e \left(\sum_j n_j Z_j^2/n_e A_j\right)^{2/3}$.
- $\beta = Z_{\mathrm{eff}}/\bar{Z}$, with $\bar{Z} = \sum_j n_j Z_j^2 (m_b/m_j)/n_e$, is the ratio of the ion pitch-angle scattering rate to the ion drag on the speed. (Written as $l(l+1)\beta'/3$ with $\beta' = Z_{\mathrm{eff}}/2\bar{Z}$ elsewhere in the literature; for the $l = 1$ moment both give the exponent $Z_{\mathrm{eff}}/3\bar{Z}$.)

The electron shielding with the trapped-electron correction (Start & Cordey 1980) gives the net driven current:

$$j_{NB} = j_f \left[1 - \frac{Z_b}{Z_{\text{eff}}}(1 - G)\right], \qquad G = \left(1.55 + \frac{0.85}{Z_{\text{eff}}}\right)\sqrt{\epsilon} - \left(0.2 + \frac{1.55}{Z_{\text{eff}}}\right)\epsilon,$$

with $\epsilon = \rho a / R_0$ and $G$ clipped to $[0, 1]$. Total currents and the current-drive efficiency:

$$I = \frac{V}{2\pi R_0}\langle j \rangle, \qquad \eta_{CD} = \frac{I_{NB}\, R_0\, \bar{n}_{e,20}}{P_{NB}} \quad [10^{20}\,\mathrm{A\,W^{-1}m^{-2}}],$$

with $P_{NB}$ the injected power and $\bar{n}_e$ the line-average density.

### 1.14.2 Bootstrap current

Sauter, Angioni & Lin-Liu, *Phys. Plasmas* **6** (1999) 2834 (erratum **9** (2002) 5140), valid at arbitrary aspect ratio and collisionality, in the cylindrical form

$$j_{BS} = -\frac{1}{B_p}\left[L_{31}\frac{dp}{dr} + L_{32}\, n_e \frac{dT_e}{dr} + L_{34}\, \alpha\, n_i \frac{dT_i}{dr}\right],$$

with $p$ the **thermal** pressure (fast ions excluded; their current is the NBCD), $L_{31}, L_{32}, L_{34}, \alpha$ the Sauter fits in the trapped fraction $f_t$ (Lin-Liu & Miller), $\nu_e^*$, $\nu_i^*$ and $Z_{\text{eff}}$, and $dr = a\, d\rho$. The profile gradients are taken analytically from the $(1 - \rho^2)^{2p}$ shapes and the current is integrated on a fine grid clustered at the edge: for low peaking exponents ($2p < 1$) the gradients diverge at $\rho = 1$, and a finite-difference gradient would make the edge contribution grid-dependent.

The 0-D model has no current profile, so $B_p$ and $q$ come from an **assumed** one:

$$j \propto (1 - \rho^2)^{\nu}, \qquad \nu = \frac{q_{cyl}}{q_0} - 1, \qquad B_p = \frac{\mu_0 I(\rho)}{L_p}, \quad L_p = 2\pi\rho a\sqrt{\frac{1 + \kappa^2}{2}},$$

with $I(\rho) = I_p[1 - (1 - \rho^2)^{\nu+1}]$, $q_{cyl}$ the cylindrical edge $q$ (`safety_factor_cyl_edge`) and $q_0$ an input (Plasma row `q0`, default 1). The low-aspect-ratio toroidal enhancement of $q$ ($q_{95}/q_{cyl}$, large on an ST) is not a current-peaking effect, so it enters $q(\rho)$ only, used in the collisionalities:

$$q(\rho) = q_{cyl}(\rho)\left[1 + \left(\frac{q_{95}}{q_{cyl}} - 1\right)\rho^2\right].$$

(Taking $\nu$ from $q_{95}$ instead gave $\nu \approx 16$ on MAST-U and NSTX-U, an unrealistically peaked current.)

The inductive remainder is $I_{ind} = I_p - I_{NB} - I_{BS}$ (negative: over-drive), and $f_{NI} = (I_{NB} + I_{BS})/I_p$.

### 1.14.3 Validation

TRANSP, ASTRA and CRONOS references (details, figures and scripts in `docs/Validation/HI-Jass_validation.md`, sec. F-H). NSTX-U, MAST and TCV use the centre-of-mass cross-sections and are given for both deposition options (cutoff / consistent), with $\tau_E$ tuned to the reference stored energy or temperatures; ITER and JET at matched $\beta_N$ (cutoff, earlier runs):

| Case | $I_{NB}$ model / ref. | $I_{BS}$ model / ref. |
|---|---|---|
| NSTX-U 0.87 MA, 10.2 MW 80 kV, H98=1 | 353 / 285 / 287 kA | 594 / 621 / 583 kA |
| NSTX-U 0.975 MA, 12.6 MW 90 kV, H98=1 | 404 / 327 / 370 kA | 653 / 682 / 604 kA |
| NSTX-U 1.1 MA, 15.6 MW 100 kV, H98=1 | 466 / 381 / 396 kA | 762 / 793 / 704 kA |
| NSTX-U 0.635 MA, 0.75 T, 6.8 MW 80 kV | 199 / 211 / 184 kA | 441 / 444 / 451 kA |
| MAST #18808 0.62 MA, 3.5 MW off-axis | 287 / 268 / 253 kA | 147-161 / $\approx$ 45 kA |
| TCV #59429 0.15 MA, 1.15 MW (ASTRA) | 91-124 / 67-89 / $\approx$ 49 kA | 74-80 / $\approx$ 81 kA |
| ITER hybrid 12 MA, $\beta_N$ 2.65 | 3.33 / 3.5 MA | 5.82 / 4.7 MA |
| ITER steady state (no ITB) 10 MA, $\beta_N$ 2.70 | 4.31 / 3.6 MA | 5.45 / 5.0 MA |
| JET #58323 1.4 MA, $\beta_N$ 2.8 | 0.55 / 0.33 MA | 0.51 / 0.32 MA |

References: NSTX-U, Gerhardt et al., *Nucl. Fusion* **52** (2012) 083020 (fully non-inductive TRANSP scenarios, Table 2, broad profiles; $W_{tot}$ matched, $f_{NI}$ = 1.03-1.07 with the consistent deposition). MAST, Turnyanskiy et al., *Nucl. Fusion* **49** (2009) 065002 (TRANSP classical; the bootstrap value is read off a stacked plot). TCV, Piron et al., *Nucl. Fusion* **59** (2019) 096012 (ASTRA profiles, digitised and integrated, about 20 % uncertainty; ECCD not modelled). ITER, Parail et al., *Nucl. Fusion* **53** (2013) 113002 (Table 3). JET, Sips, EPS 2003 O-1.3A (TRANSP; density and beam set-up only approximately known, $a = 0.95$ m used for the diverted plasma).

With the consistent deposition NBCD agrees within about 15 % on NSTX-U and 6 % on MAST; TCV stays 1.4-1.8x high (charge-exchange loss the main uncertainty). The cutoff deposition overpredicts NBCD wherever the orbits are wide. The bootstrap current agrees within about 15 % on NSTX-U and TCV and 25 % at ITER, but is about 3x the MAST reference and high on JET. Matching $\beta_N$ instead of the stored energy raises both NSTX-U currents by 20-30 % (the codes normalise $\beta_N$ differently), so cross-code comparisons should match $W$. Remaining systematic effects: the **full-energy-only beam** ($E/2$ and $E/3$ not modelled; on NSTX-U they lowered $I_{NB}$ by about 15 % in an earlier test) and a first-orbit loss of 25-40 % on spherical tokamaks, well above TRANSP's few per cent.

### 1.14.4 Limitations

- Each energy component ($E$, $E/2$, $E/3$) is a separate mono-energetic source; no anomalous fast-ion diffusion, no current diffusion (steady state assumed).
- Assumed current profile and $q(\rho)$; no equilibrium.
- Where fast ions make the thermal-ion density hollow (fixed $n_e$, large $n_b$ in the core), the thermal bootstrap reverses near the axis (seen on DANTE).
- Low-peaking presets ($p \simeq 0.1$ to 0.25) put roughly 28 % of $I_{BS}$ at $\rho > 0.95$: the power-law shapes act like a pedestal.

## 1.15 Confinement time

$\tau_{E,e}$ and $\tau_{E,i}$ in the balances above are set per channel by the *Confinement* selector in the Losses section; the electron and ion channels are chosen independently. When a physics-based scaling is selected the manual `tauE,e` / `tauE,i` inputs are ignored, and the Summary sheet reports the computed value (or its range across a scan) instead.

The physics-based scalings below all use the same loss power

$$P_{\text{loss}} = \sum_j P_{\text{useful},j} + P_{\text{aux},e} + P_{\text{aux},i},$$

i.e. the captured, post-orbit, post-charge-exchange beam power plus the prescribed auxiliary (ECRH, ICRH) heating. Fusion alpha heating is not included in $P_{\text{loss}}$ (it is closed by an outer

fixed-point and is not known when $\tau_E$ is evaluated). None of these depend on $T_e$ or $T_i$, so each is evaluated once.

### 1.15.1 Confinement enhancement

Every scaling below can be multiplied by a confinement enhancement $h$ (default 0; the MODELS slider under Confinement in the HOT-Jass web app, and the config field `tauE_enhancement` in HI-Jass):

$$\tau_E = (1 + h)\,\tau_{\text{scaling}}, \qquad H = 1 + h.$$

$h = 0$ is the plain scaling. Fixed $\tau_E$ input and the neoclassical ion channel are not affected.

### 1.15.2 Fixed (input)

$\tau_{E,e}$ and $\tau_{E,i}$ are the values entered in the Losses section (under the Confinement selector), used unchanged.

### 1.15.3 IPB98(y,2) ELMy H-mode

The ITER Physics Basis ELMy H-mode scaling (ITER Physics Basis, *Nucl. Fusion* **39** (1999) 2175; standard reference form):

$$\tau_E = 0.0562\, I_p^{0.93}\, B_t^{0.15}\, n_{19}^{0.41}\, P_{\text{loss}}^{-0.69}\, R_0^{1.97}\, \kappa_a^{0.78}\, \varepsilon^{0.58}\, M_{\text{eff}}^{0.19}$$

with $I_p$ in MA, $B_t$ in T, $P_{\text{loss}}$ in MW, $R_0$ and $a$ in m, $n_{19} = n_{e0}/10^{19}\,\text{m}^{-3}$, $\varepsilon = a/R_0$, $\kappa_a = \kappa$, and $M_{\text{eff}} = 2x_D + 3x_T$ the mass-weighted ion mass number. Selecting *IPB98(y,2) ELMy H-mode* uses it for both channels ($\tau_{E,e} = \tau_{E,i}$) – like every scaling offered here, it does not split electron/ion channels.

The fit was obtained from conventional-aspect-ratio devices ($A \sim$ 2.5-4) and is documented in the spherical-tokamak literature to mis-predict at low aspect ratio. When $A = R_0/a < 2$ the Assumptions tab flags that IPB98(y,2) is being extrapolated outside its dataset.

### 1.15.4 Kaye NSTX L-mode

An NSTX low-aspect-ratio (spherical tokamak) L-mode global fit (Kaye, *Nucl. Fusion* **46** (2006) 848, Eq. 5 / Fig. 9):

$$\tau_E = 4.73 \times 10^{-4}\, I_p^{1.01}\, B_t^{0.70}\, n_e^{-0.07}\, P_{\text{loss}}^{-0.37}$$

with $I_p$ in A, $B_t$ in T, $n_e$ in $\text{m}^{-3}$, $P_{\text{loss}}$ in W and $\tau_E$ in s. Selecting *Kaye NSTX L-mode* uses it for both channels ($\tau_{E,e} = \tau_{E,i}$).

### 1.15.5 Kaye NSTX H-mode

The H-mode counterpart of the previous fit: an NSTX low-aspect-ratio (spherical tokamak) H-mode thermal-energy scaling (Kaye, *Nucl. Fusion* **46** (2006) 848, Table 1, ordinary-least-squares "Case 1" fit to all 85 H-mode points, RMSE = 0.145):

$$\tau_E = 4.69 \times 10^{-9}\, I_p^{0.57}\, B_t^{1.08}\, n_e^{0.44}\, P_{\rm loss}^{-0.73}$$

with $I_p$ in A, $B_t$ in T, $n_e$ in m$^{-3}$, $P_{\rm loss}$ in W and $\tau_E$ in s. Relative to IPB98(y,2) the current dependence is much weaker ($I_p^{0.57}$ vs $I_p^{0.93}$) and the toroidal-field dependence much stronger ($B_t^{1.08}$ vs $B_t^{0.15}$), the qualitative spherical-tokamak trend; the density and heating-power exponents ($n_e^{0.44}$, $P_{\rm loss}^{-0.73}$) are of the same order as the conventional-aspect scalings. This is an ST-appropriate fit ($A \sim 1.3$ to 1.5 dataset). Selecting *Kaye NSTX H-mode* uses it for both channels ($\tau_{E,e} = \tau_{E,i}$).

### 1.15.6 Neoclassical ion

**Not currently selectable from the Confinement dropdown** – v1 offers only the four modes above (Fixed, Kaye L, Kaye H, IPB98(y,2)), each applied to both channels. The neoclassical estimate below is implemented in `hotjass/solve.py` and was previously reachable via an "X e / neoclassical i" hybrid selector entry; that combination is out of scope for v1 and may return in a later version. Documented here for completeness:

A banana-regime order-of-magnitude estimate (Wesson, *Tokamaks*; Chang-Hinton low-collisionality limit) for the ion channel only:

$$\chi_{i,\rm neo} \simeq \frac{q^2\, \rho_i^2\, \nu_{ii}}{\varepsilon^{3/2}}, \qquad \tau_{E,i} = \frac{a^2}{\chi_{i,\rm neo}} = \frac{a^2\, \varepsilon^{3/2}}{q^2\, \rho_i^2\, \nu_{ii}},$$

with $\varepsilon = a/R_0$, the elongation-corrected cylindrical safety factor

$$q = \frac{5\, a^2 B_t}{R_0\, I_p[\rm MA]}\, \frac{1+\kappa^2}{2},$$

the thermal-ion Larmor radius

$$\rho_i = \frac{m_{\rm eff} v_{th,i}}{e B_t}, \qquad v_{th,i} = \sqrt{\frac{2 T_i (10^3 e)}{m_{\rm eff}}}, \qquad m_{\rm eff} = x_D m_D + x_T m_T,$$

and the ion-ion collision frequency summed over D and T,

$$\nu_{ii} = \sum_{s \in \{D,T\}} 4.80 \times 10^{-8}\, \frac{n_s[\rm cm^{-3}]\, \ln\Lambda}{\sqrt{\mu_s}\, T_i[\rm eV]^{3/2}}, \qquad \mu_s = m_s/m_p.$$

Since $\tau_{E,i} \propto T_i^{1/2}$, the ion balance becomes a 1-D fixed-point root-find for $T_i$ rather than a closed-form substitution. Pure neoclassical transport has no anomalous channel, so it under-estimates real ion transport and can predict very hot, sometimes implausible $T_i$; treat it as a lower bound. It cannot be combined with electron-ion equipartition.

### 1.15.7 Neoclassical ion (arbitrary aspect ratio)

**Also not currently selectable** – same v1 scope note as above.

The bare $\varepsilon^{-3/2}$ geometric factor above both diverges as $\varepsilon \to 0$ and stays bounded as $\varepsilon \to 1$, neither of which is right on a spherical tokamak. The *arbitrary A* variant (previously, selector entries ending "*neoclassical i (arbitrary A)*") replaces the trapped-fraction part with the Lin-Liu

& Miller $f_t/f_c$ the banana-regime structure of Helander & Sigmar's *Collisional Transport in Magnetized Plasmas*, Ch. 11 and uses the arbitrary-$A$ edge safety factor $q_a$:

$$\chi_{i,\mathrm{neo}} \simeq q_a^2\,\rho_i^2\,\nu_{ii}\,\frac{f_t}{f_c}, \qquad f_t = 1 - \frac{(1-\varepsilon)^2}{(1+1.46\sqrt{\varepsilon})\sqrt{1-\varepsilon^2}}, \qquad f_c = 1 - f_t,$$

$$q_a = \frac{5\,a^2 B_t}{R_0\,I_p[\mathrm{MA}]}\,\frac{1+\kappa^2(1+2\delta^2-1.2\delta^3)}{2}\,\frac{1.17-0.65\,\varepsilon}{(1-\varepsilon^2)^2}.$$

$f_t/f_c$ stays finite for every $\varepsilon < 1$ and rises steeply only as $\varepsilon \to 1$ (all ions trapped) the physically correct low-aspect trend. This is **not** a smooth reduction of the $\varepsilon^{-3/2}$ form: the two order-of-magnitude estimates can differ by up to about one order of magnitude in either direction (more once the $\tau_{E,i} \propto T_i^{1/2}$ fixed point amplifies it), which is exactly why both are offered run them side by side. References: Helander, *Phys. Plasmas* **7** (2000) 3999; Hinton, Wiley *et al.*, *Phys. Rev. Lett.* **29** (1972) 698; Satake *et al.*, *Phys. Plasmas* **9** (2002); Goldston & Rutherford (1995).

### 1.15.8 Known limitation: the $P$ fed into IPB98(y,2) / Kaye NSTX

These three scalings use $P_{\mathrm{loss}} = \sum_j P_{\mathrm{useful},j} + P_{\mathrm{aux},e} + P_{\mathrm{aux},i}$ (sec. above) NBI power *after* shine-through, first-orbit loss **and** charge-exchange loss are all subtracted. The experimental databases these scalings were fit to use a coarser convention instead (absorbed power, i.e. shine-through-corrected only they do not subtract orbit-loss/CX-loss of the already-absorbed fast-ion population), so HI-Jass currently under-feeds $P$ into these negative-exponent scalings and is biased toward *overestimating* $\tau_E$ whenever orbit-loss/CX-loss are significant (STs especially). Not yet fixed. See **Appendix A** for a quantitative comparison, including how this compares to the (much larger) effect of the scaling choice itself.

## 1.16 Mean fast-ion energy

Each beam has its own steady-state slowing-down distribution:

$$f_j(E) \propto \frac{\sqrt{E}}{E^{3/2} + E_{c,j}^{3/2}}, \qquad 0 < E \le E_{b,j}$$

Its mean fast-ion energy is:

$$\langle E_{fast} \rangle_j = \frac{\int_0^{E_{b,j}} E f_j(E)\,dE}{\int_0^{E_{b,j}} f_j(E)\,dE}$$

## 1.17 Combined values shown in Results

The GUI reports power-weighted effective values for the two beams.

Effective slowing-down time:

$$\tau_S^{\mathrm{eff}} = \frac{\sum_j P_{\mathrm{useful},j}\tau_{S,j}}{\sum_j P_{\mathrm{useful},j}}$$

Effective mean fast-ion energy:

$$\langle E_{fast}\rangle = \frac{\sum_j P_{\mathrm{useful},j}\langle E_{fast}\rangle_j}{\sum_j P_{\mathrm{useful},j}}$$

These effective values are used for the combined fast-ion diagnostics, including fast-ion energy density, pressure, beta, and:

$$R = \frac{u_{fast}}{U_t}$$

The underlying HotJass calculation still evaluates each beam separately before forming these combined diagnostics.

### 1.18 Plasma beta: isotropic and anisotropic fast ions

Toroidal and normalised beta use the volume-averaged pressure of thermal electrons, thermal ions and fast beam ions:

$$\beta_t = \frac{\langle p\rangle}{B_0^2/2\mu_0}, \qquad \beta_N = \beta_t[\%]\,\frac{a\,B_0}{I_p[\mathrm{MA}]}, \qquad \langle p\rangle = \langle n_e T_e\rangle + \langle n_i T_i\rangle + p_{fast}.$$

The fast-ion pressure depends on the pitch distribution, which the 0-D model does not resolve. With $u_{fast} = n_b\langle E_{fast}\rangle$ the fast-ion energy density and $\xi = v_\parallel/v$:

$$p_{fast} = \left(1 - \langle\xi^2\rangle\right) u_{fast}.$$

- **Isotropic** ($\langle\xi^2\rangle = 1/3$, the default): $p_{fast} = \frac{2}{3}u_{fast}$.
- **Anisotropic** bound ($\langle\xi^2\rangle = 1$, all fast ions parallel): $p_{fast} = 0$.

The solver reports the first as `beta_t` (pitch set by `fast_ion_mean_pitch2`) and always computes the second as `beta_t_anisotropic`.

Both are shown (the anisotropic value in brackets). Tangential beams are born nearly parallel and isotropise only by pitch-angle scattering while slowing down, so the true value lies between the two bounds: closer to the anisotropic one for tangential injection at high $T_e$ (slowing down faster than scattering), closer to the isotropic one for perpendicular beams. The difference matters where fast ions carry a large share of the pressure (low density, high beam power). $\beta$ is a diagnostic: nothing in the solve depends on it.

**Fast-ion pressure in the NBI_OUT export (HOT-Jass web).** The parallel and perpendicular pressures written to NBI_OUT (PBLON, PBPER) follow the *Fast-ion pressure* choice at the end of MODELS:

$$p_\parallel = 2\langle\xi^2\rangle\,u_{fast}, \qquad p_\perp = \left(1 - \langle\xi^2\rangle\right) u_{fast}, \qquad p_\perp + p_\parallel/2 = u_{fast},$$

so the stored energy does not depend on it. *Isotropic* (1/3) and *Anisotropic (parallel)* (1) are the two bounds above. The default, *Birth pitch + scattering*, takes a local $\langle\xi^2\rangle(\rho)$ per beam component from the steady slowing-down distribution of the NBCD model (electron and ion drag plus ion pitch-angle scattering, see *Neutral-beam current drive*). The $l = 2$ Legendre moment of the pitch starts at the birth value and decays as $G^\beta$ while the ion slows down ($G^{\beta/3}$ is the $l = 1$ factor of the NBCD integral); weighting by energy over the distribution $\propto u^2/(u^3 + u_c^3)$:

$$\langle \xi^2 \rangle = \frac{1}{3} + \frac{2}{3} P_2(\xi_0) \frac{\int_0^1 \frac{u^4}{u^3+u_c^3} G^\beta \, du}{\int_0^1 \frac{u^4}{u^3+u_c^3} \, du}, \qquad G = \frac{u^3(1+u_c^3)}{u^3+u_c^3}, \qquad P_2 = \frac{3\xi_0^2 - 1}{2},$$

with $u = v/v_b$, $u_c$, $\beta = Z_{eff}/\bar{Z}$ and the profiles exactly as in the NBCD calculation and $\xi_0(\rho)$ the birth-weighted mean pitch (the pitch spread within a radial bin is neglected). Beam-averaged values on the presets: NSTX-U 0.23 / 0.49 (NBI-1 is born at low pitch, more perpendicular than isotropic), MAST-U 0.50 / 0.57, ITER 0.49 / 0.55 – between the bounds, as expected for tangential beams.

### 1.19 Results tabs (Operating point)

- **Dashboard** scalar table of the operating point (temperatures, and their on-axis values when profile-corrected 0-D is on; densities, power balance terms, orbit widths, D-T / D-D fusion power, neutron rate, Greenwald ratio, edge safety factor $q_*$, beta). $q_*$ is reported using whichever formula the active orbit-loss model actually uses – the bare cylindrical `safety_factor_cyl_edge` for "large-aspect", or the arbitrary-$A$ `safety_factor_cyl_edge_arbitrary_A` ($q_a$) for either ST orbit variant – so it stays consistent with the orbit widths shown just above it. Per-beam CX-loss rows show $f_{\mathrm{cx},P}/f_{\mathrm{cx},N}/\gamma_{\mathrm{cx}}$ (plus $n_0$ and the informational escape probability) when a physics-based CX model is selected, or just the flat fraction for "Manual fraction" – see "Charge-exchange loss" above. The rotation row shows "off" unless a Rotation model is selected, in which case it shows $v_\phi$, a Mach number, and (momentum-balance mode) the net NBI torque – see "Bulk toroidal rotation" above.
- **Deposition** (1) beam targeting geometry in the torus top view (titled with the active device), tangent to each beam's $R_t$ with its co-/counter-current sense; (2) neutral-beam survival $I(s)/I_0$ and the fast-ion birth rate $n_e\sigma\, I(s)/I_0$ along the beam, annotated with each beam's shine-through percentage; (3) the resulting fast-ion birth versus normalised radius $\rho$, per beam and total (chord-sampled, midplane circular approximation), with the prompt first-orbit-loss zone $\rho > \rho_{cut}$ shaded and a per-beam $\rho_{cut}$ line (from the same criteria as the orbit-loss model in use); (4) the steady-state slowing-down distribution $f(E)$ with the $E_b$ edge marked.
- **Power flow** waterfall / Sankey / pie of injected power to its sinks.
- **Profiles** $n_e(\rho)$, $T_{e,i}(\rho)$, $\tau_S(\rho)$ (local thermalization time, from the same local $n_e(\rho)$, $T_e(\rho)$ used elsewhere in this panel – log scale, since it rises sharply toward the edge as $n_e \to 0$), $n_0(\rho)$ (background-neutral density behind the active CX-loss model – see "Charge-exchange loss" above), the plasma shape, and $P_{fus}(\rho)$ (local D-T + D-D fusion power density, thermal vs. beam-plasma / beam-target, built from the same on-axis densities and $(1-\rho^2)^{2p}$ shapes as the volume-integrated totals on the Dashboard – see "Fusion power" below).
- **Fusion** a pie chart of $P_{fus}$ split into its six channel/reaction combinations (D-T and D-D, each thermal / beam-target / beam-beam), with MW values and percentages; slices under 0.5% of the total are hidden to keep the chart readable (still fully available in the Dashboard's D-T/D-D breakdown rows). Title also shows $Q$ and the full neutron rate $Y_n$.
- **Assumptions** the full validity / fit-range read-out and any warnings.
- **References** the literature behind the *currently selected* models (beam stopping, confinement scaling, orbit-loss model) and the active machine geometry, each with a link, plus a feedback button.

### 1.20 Saving and loading a run

Below the results tabs, **Export** has three buttons plus a fourth for the reverse direction:

- **Summary PNG / Summary PDF** a snapshot of the Summary tab (the fixed 3x4 scan-panel grid) as an image.
- **Run record (JSON)** every plasma/beam input, every model toggle, the full solved operating point, and the Summary sheet's scan arrays, as one JSON file (needs a completed run; disabled with a status message otherwise).
- **Load run record...** the reverse of the above: pick a previously saved run-record JSON, and every input field and toggle (confinement, orbit loss, CX-loss model, rotation, beam-beam, equipartition, alpha self-heating, profile-corrected 0-D, both NBI sections) is restored from it, then the point is re-solved automatically. Tolerant of a record written by an older or newer HI-Jass version: a field this version doesn't recognise is ignored, and a field this version expects but the record lacks simply keeps its current value instead of raising an error. Only the solved *inputs* are restored – the record's own `operating_point /summary` output is historical reference only, not re-applied (the fresh re-solve after loading is what actually reproduces those numbers, so loading a record from a much older HI-Jass version may reproduce a slightly different result if the underlying physics changed meanwhile).

### 1.20.1 U-files for ASTRA / TRANSP (HOT-Jass web)

In HOT-Jass web, **Results → Save → Save U-files** writes the last calculation as ASCII U-files (the TRANSP / ITPA profile-database format read by ASTRA and other transport codes): one quantity per file, all in one zip named `hot_jass_ufiles_<TOK><shot>.zip`, with a `README.txt` listing every file. The shot number is set in the dialog (*Shot #*); the 4-character tokamak tag comes from the active device name. Files are named `<TOK><shot>_<NAME>.2d` / `.1d`.

The operating point is steady state, so every file carries the same values at two times, $t = 0$ and 100 s: a code reading any time in between gets this point. Radial profiles use 51 points in $\rho$ from 0 to 1, where $\rho$ is the HOT-Jass flux label $\sqrt{((R - R_0)/a)^2 + (Z/\kappa a)^2}$ (normalised minor radius on elliptical surfaces), **not** the toroidal-flux $\rho_{tor}$.

2-D files, $f(\rho, t)$:

| Name | Units | Quantity |
|---|---|---|
| NE | M**-3 | electron density |
| TE | EV | electron temperature |
| TI | EV | ion temperature |
| ZEFF | – | effective charge (flat) |
| NFAST | M**-3 | NBI fast-ion density |
| QNBIE | W/M**3 | NBI heating of electrons |
| QNBII | W/M**3 | NBI heating of ions |
| SNBII | M**-3/S | NBI fast-ion source (confined births) |
| PFUS | W/M**3 | fusion power density (thermal + beam-target) |
| CURNBI | A/M**2 | NB-driven current density |
| CURBS | A/M**2 | bootstrap current density |
| Q | – | safety factor of the assumed current profile |

1-D files, $f(t)$: IP [A] and BT [T] (signed, positive counter-clockwise seen from above), PNBI [W] (injected NBI power), RGEO and AMIN [M] ($R_0$, $a$), KAPPA, DELTA.

Notes:

- QNBIE, QNBII, SNBII, PFUS, CURNBI, CURBS and Q need profile-corrected 0-D (MODELS); without it only the other files are written, and the README says so.
- QNBIE / QNBII put the deposited NBI power at the birth radius and split it between electrons and ions with each beam component's slowing-down fraction (no radial redistribution

during slowing down). The README checks $\int (Q_{NBIE}+Q_{NBII})\,dV$ against the useful NBI power of the power balance and warns if they differ: with the historical cutoff deposition, a beam born entirely outside the orbit-loss cutoff radius has no births although its useful power is counted ("Orbit loss consistent in deposition" or the $P_\phi$ orbit model avoid this).
- CURNBI, CURBS and Q are diagnostics on an assumed current profile (Current tab), not a current evolution.
- Format: label lines with `;-...-` descriptions from column 32 (name 20 characters, units 10), then the X grid, the time grid and $f$ with X varying fastest, 6 numbers per line, 13 characters each (`E13.5`, so a negative value keeps a leading blank and both fixed-width and whitespace readers work), then the `;----END-OF-DATA` trailer and free-text comments.

### 1.20.2 NBI_OUT for ASTRA (HOT-Jass web)

**Results → Save → NBI_OUT** writes `nbi_out.dat` in the NBI_OUT v1 format of the file-based NBI coupling to ASTRA (module → ASTRA): line 1 `# NBI_OUT v1`; 15 scalar `KEY value` lines in fixed order; `SOURCES` and one row per beamline (source number, PINJ, PSHINE, PION, PLOSS [W], with the $E$, $E/2$, $E/3$ components added); `NPROF 22`, a `COLUMNS` line and NA1 rows; `END`. SI units ($m^{-3}$, W, A, Pa, N); reals `ES24.16`, integers `I10`, no comment lines. **NBI_OUT Preview** plots all 21 columns against RHO with the scalars before saving; both use the same data.

HOT-Jass is not called by an ASTRA stub (no NBI_IN), so TIME = NCALL = 0. The grid is NA1 cell centres, set in the dialog (default 100), $\rho_j = (j-\frac{1}{2})/N_{A1}$ of the HOT-Jass elliptical label, written as RHO $= \rho_j\, a\sqrt{\kappa}$ [m] (approximately the toroidal-flux radius $\sqrt{\Phi/\pi B_0}$; not ASTRA's own equilibrium $\rho$). Volume integrals use $\int f\,dV = V\sum_j f_j\, 2\rho_j \Delta\rho$.

Profile columns, from the solver's profiles and the confined birth rate $S_k h_k(\rho)$ of each energy component $k$ (normalised to its useful power):

| Column | Units | Computed as |
|---|---|---|
| RHO | m | $\rho_j a\sqrt{\kappa}$ |
| PEBM, PIBM | $W/m^3$ | $\sum_k L_{e,k} S_k h_k E_k$, $\sum_k L_{i,k} S_k h_k E_k$ |
| PBEAM | $W/m^3$ | PEBM + PIBM |
| NIBM | $m^{-3}$ | $\sum_k S_k h_k\, \tau_{th,k}(n_e, T_e)$ |
| PBLON, PBPER | Pa | $2\langle\xi^2\rangle u$, $(1-\langle\xi^2\rangle)\,u$ |
| CUFI, CUBM | $A/m^2$ | Current tab: unshielded, shielded |
| SNEBM | $m^{-3}s^{-1}$ | $\sum_k S_k h_k$ |
| SNNBM | $m^{-3}s^{-1}$ | zeros (not modelled) |
| SNIBM1, 2, 3 | $m^{-3}s^{-1}$ | $S_k h_k$ of the H, D, T components |
| SCUBM | $N/m^3$ | $\pm\sum_k S_k h_k\, m_b v_b R_t/R_0$ |
| NNBM1, 2, 3 | $m^{-3}$ | zeros (not modelled) |
| STNBDP, SDNBDP1/2, SDNBTP | $s^{-1}$ | $n_b\langle\sigma v\rangle$ per target ion |

PEBM and PIBM put the deposited power at the birth radius and split it with each component's slowing-down fractions $L_{e,k}$, $L_{i,k}$; PBEAM counts confined births only. NIBM is the solver's fast-ion density. In PBLON and PBPER, $u = \sum_k n_{b,k}\langle E\rangle_k$ and $\langle\xi^2\rangle$ follows *Fast-ion pressure* (see *Plasma beta*). SCUBM is positive along $I_p$. The beam-target rates use the local slowing-down spectrum: STNBDP T beam on D; SDNBDP1 D on D, t + p branch; SDNBDP2 D on D, n + $^3$He branch; SDNBTP D beam on T.

Scalars: PINJ, PSHINE, PION, PLOSS (prompt orbit + CX) from the solver's power balance; PHEAT $= \int(\text{PEBM}+\text{PIBM})\,dV$; WFAST $= \int(\text{PBPER}+\text{PBLON}/2)\,dV$; IBM the NB-driven current of the Current tab; TORQ $= \sum 2P_{useful}R_t/v_b$ (also with rotation off); NRATE the beam-target plus beam-beam neutron rate; NSRC the number of beamlines. STATUS: 0 OK; 1 if

$|\mathrm{PION} - \mathrm{PHEAT} - \mathrm{PLOSS}| > 10^{-2}\,\mathrm{PION}$, with the difference in MESSAGE (the historical cutoff deposition, or a coarse grid: below about 50 points the grid sum is off by 1–2 %); −3 without profile-corrected 0-D (all columns but RHO zero).

Checks on the presets: PION = PHEAT + PLOSS within 0.1 %; NIBM within 0.5 % of the solver's fast-ion density; $R_0 \int \mathrm{SCUBM}\, dV = \mathrm{TORQ}$; the beam-target columns times the thermal D, T densities reproduce the solver's beam neutron rate. Approximations: global $L_e$, $L_i$ per component and no radial redistribution while slowing down; ions lost promptly or by CX are not in the sources; SCUBM uses $R_0$ for the local $R$; the SDNBDP1/2 branch assignment follows the v1 specification, which itself lists it as still to be checked against ASTRA.

### 1.21 Interface font size (accessibility)

The  button next to the Operating point/Scan mode toggle, above the Run button, cycles the whole interface through four zoom levels: 100% → 115% → 130% → 150% → back to 100%. Each click rescales every rail label/entry/button, the Dashboard, and all plot text (titles, axis labels, legends, tick labels) together, redrawing the currently solved point immediately so nothing looks mismatched. The chosen level is written to `~/.hi_jass/settings.json` and restored automatically on the next launch.

### 1.22 Electron-ion exchange time

The electron-ion exchange time is based on the NRL Plasma Formulary thermal-equilibration rate. For each thermal ion species $s$ (D or T):

$$\bar{\nu}_{ei,s} = 1.8 \times 10^{-19} \frac{\sqrt{m_e m_s}\, Z_e^2 Z_s^2\, n_e \ln\Lambda}{(m_s T_e + m_e T_i)^{3/2}}$$

In this practical-unit expression, masses are in grams, temperatures are in eV, and electron density is in $\mathrm{cm}^{-3}$. The model uses $Z_e = Z_s = 1$ for hydrogenic D and T ions.

For a D-T plasma, the effective exchange time is:

$$_{IE} = \frac{n_D + n_T}{n_D \bar{\nu}_{ei,D} + n_T \bar{\nu}_{ei,T}}$$

For a single thermal ion species, this reduces to:

$$_{IE} = \frac{1}{\bar{\nu}_{ei}}$$

The equivalent energy-over-power form used by the application is:

$$_{IE} = \frac{\frac{3}{2}(n_D + n_T)|T_i - T_e|(10^3 e)V}{|P_{ie}|}$$

where $V$ is the plasma volume and $e = 1.602176634 \times 10^{-19}$ J/eV. The exchange power is:

$$P_{ie} = -P_{ei} = -\frac{3}{2} V (T_e - T_i)\,(n_D \bar{\nu}_{ei,D} + n_T \bar{\nu}_{ei,T})\,(10^3 e)$$

The absolute values make $\tau_{IE}$ positive regardless of whether electrons heat ions ($T_e > T_i$) or ions heat electrons ($T_i > T_e$).

## 1.23 Fusion power

The total fusion power is the sum of the thermal D-T and beam-target contributions:

$$P_f = P_{f,\text{thermal}} + P_{f,\text{beam}}$$

### 1.23.1 Thermal D-T fusion

The thermal contribution is calculated from the thermal deuterium and tritium densities and the ion-temperature-dependent Bosch-Hale reactivity:

$$P_{f,\text{thermal}} = n_{D0} n_{T0} \langle \sigma v \rangle_{DT}(T_i) V E_f$$

Here $n_{D0}$ and $n_{T0}$ are the thermal species densities, $V$ is the plasma volume, and:

$$E_f = 17.6\,\text{MeV} = 17.6 \times 10^6 e$$

The Bosch-Hale thermal reactivity is evaluated at $T_i$, because the relative velocity is that of the thermal ions, not the electrons.

### 1.23.2 Beam-target fusion

Each useful beam is evaluated separately against the stationary thermal ions of the other D-T species:

$$P_{f,\text{beam}} = \sum_j V E_f \int_0^{E_{b,j}} n_{\text{target},j} f_j(E) [\sigma v]_j(E)\, dE$$

The beam slowing-down distribution is normalized to the fast-ion density of that beam:

$$f_j(E) \propto \frac{\sqrt{E}}{E^{3/2} + E_{c,j}^{3/2}}, \qquad \int_0^{E_{b,j}} f_j(E)\, dE = n_{b0,j}$$

The beam-target reactivity is:

$$[\sigma v]_j(E) = \sigma_{DT}(E) v_j(E), \qquad v_j(E) = \sqrt{\frac{2E}{m_j}}$$

**Cross-section energy.** The Bosch-Hale cross-section fits take the centre-of-mass energy, while every beam-target and beam-beam call passes the lab energy of a deuteron on a target at rest, $\frac{1}{2} m_D v_{rel}^2$. By default (historical) the fits are evaluated at that lab energy. With `cm_energy_correction` (HOT-Jass: MODELS, "COM energy in beam cross-sections") they use

$$E_{cm} = E \frac{m_{target}}{m_D + m_{target}} : \qquad E_{cm} = E/2 \text{ (D-D)}, \qquad E_{cm} \simeq 0.6\, E \text{ (D-T)}.$$

The lab-energy default over-predicts D-D beam-target rates (about 3x at 60 keV; MAST neutron rates 2-3.5x TRANSP instead of 0.5-0.9x) and misplaces the D-T resonance (beam-target D-T too low above about 80 keV, x0.5 at 120 keV). Thermal reactivities are not affected. Off by default until checked on more cases.

For a D-T beam-target reaction $E_f = 17.6$ MeV. For a D beam, $n_{\mathrm{target},j} = n_{T0}$; for a T beam, $n_{\mathrm{target},j} = n_{D0}$. The beam fast-ion density is obtained from the useful beam power:

$$n_{b0,j} = \frac{P_{\mathrm{useful},j}\tau_{S,j}}{E_{b,j}(10^3 e)V}$$

The input shine-through fraction is used to determine captured beam power; the plotted $P_{\mathrm{shine}}$ is then the derived shine-through power from the scan. Charge-exchange and other configured losses similarly reduce the useful power before calculating $n_{b0,j}$.

The fusion post-processing always uses the $(1-\rho^2)$ profile weighting: the thermal contribution is

$$P_{f,\mathrm{thermal}} = V E_f \left\langle n_D(\rho) n_T(\rho) \langle \sigma v \rangle_{DT}[T_i(\rho)] \right\rangle_V ,$$

and the beam-target contribution uses the local target density and the local slowing-down distribution evaluated with $T_e(\rho)$. The $n_{D0}$, $n_{T0}$, $T_e$, $T_i$ used here are the **on-axis** values (see *Profile-corrected 0-D*).

### 1.23.3 D-D fusion

The D-D channel (present whenever the deuterium fraction $x_D > 0$) is evaluated with the Bosch-Hale D-D fits, **branch-resolved**:

| branch | products | $E_f$ | neutron |
|---|---|---|---|
| D(d,p)T | T (1.01) + p (3.02 MeV) | 4.033 MeV | none |
| D(d,n)ş He | ş He (0.82) + n (2.45 MeV) | 3.269 MeV | 2.45 MeV |

**Thermal:**

$$P^{\mathrm{th}}_{f,\mathrm{DD}} = V\left[E_{ddp}\left\langle \tfrac{1}{2} n_D^2 \langle \sigma v \rangle_{ddp}(T_i)\right\rangle_V + E_{ddn}\left\langle \tfrac{1}{2} n_D^2 \langle \sigma v \rangle_{ddn}(T_i)\right\rangle_V\right],$$

with the $\frac{1}{2}$ identical-particle factor and the same $(1-\rho^2)$ profile weighting as the D-T thermal term.

**Beam-target** (a fast D ion on the thermal-D population **no** $\frac{1}{2}$ factor, the two populations are distinct):

$$P^{\mathrm{beam}}_{f,\mathrm{DD}} = \sum_{j\in \mathrm{D\ beams}} V \int_0^{E_{b,j}} n_{D0}\, f_j(E)\left[E_{ddp}\,\sigma_{ddp}(E) + E_{ddn}\,\sigma_{ddn}(E)\right] v_j(E)\, dE.$$

The cross-sections take the **lab-frame** deuteron energy (same convention as $\sigma_{DT}$). D-D does **not** produce alphas, so the alpha self-heating fixed-point closes on the D-T power only ($P_\alpha = f_\alpha(3.5/17.6)P_{f,DT}$).

### 1.23.4 Neutron rate

$$Y_n = \frac{P_{f,DT}}{17.6\,\mathrm{MeV}} \;+\; R^{\mathrm{th}}_{ddn} + R^{\mathrm{beam}}_{ddn}\ (+Y_n^{\mathrm{bb}}),$$

i.e. one 14.06 MeV neutron per D-T reaction plus the 2.45 MeV neutrons from the D(d,n) branch, plus the beam-beam contribution when that's enabled (see "Beam-beam fusion" above). Named $Y_n$, not $R_n$, to avoid confusion with the fast/thermal energy-density ratio $R$ reported elsewhere. Reported on the Dashboard, broken down by channel (beam-target / thermal / beam-beam / full)

in the Scan tab's `Y_beam, Y_thermal, Y_bb, Y_full` group and the Summary tab's "Neutron rate" panel, and as a slice of the Fusion tab's pie chart.

### 1.23.5 Current model limitations

- The beam-target integral uses stationary target ions, so the solved $T_i$ is not included in that reaction integral, and the beam is treated as a single full-energy component (no half/third fractions).
- The tritium and ş$^3$He bred by D-D are not burned further ("cat-DD" is neglected).
- The radial volume weighting uses the model's elliptical flux-surface approximation and does not yet solve a 2D equilibrium.

## 1.24 Density scan

The Plasma tab defines the requested scan range with `n_e_min` and `n_e_max`. The Results tab reports the physically valid interval separately. Points below the HotJass charge-neutrality feasibility limit are not assigned physical output values and are omitted from plots.

The $T_e, T_i$ scan plot's y-axis top is $\min(\max(T_e, T_i), 100\,\text{keV})$ – the actual data maximum, unless it exceeds 100 keV (the upper edge of the Bosch-Hale fits' validated range), in which case the axis is capped there so one runaway hot-ion point doesn't wash out the rest of the curve. The bottom is fixed at 0 (not autoscaled) so a fixed top never stretches the axis into an apparent negative margin.

The $n_D, n_T, n_{b0}$ scan group's otherwise-empty 4th panel shows the fast-ion / target-ion density ratio $n_{b0}/n_{\text{Target}}$, where $n_{\text{Target}}$ is whichever thermal D-T species the *dominant* (by injected power) NBI beam reacts with in the beam-target channel – $n_T$ for a D beam, $n_D$ for a T beam.

Four further scan groups:

- $P_e, P_i, P_{ie}, Q$ ("Powers") – $Q = P_{f,tot}/P_{NB}$ replaces the flat shine-through power here (moved to "Losses" below, where it belongs with the other loss channels).
- $P_{f,tot}, P_{f,th}, P_{f,b}, P_{f,bb}$ ("Pf") – the previously-empty 4th panel now shows $P_{f,bb}$ (Sec. "Beam-beam fusion" above – a flat zero while that's off), alongside the thermal and beam-target D-T/D-D totals; $P_{\text{useful}}$ is still available from the Dashboard.
- $P_{\text{shine}}, P_{\text{orbit}}, P_{\text{cx}}, P_{\text{lost}}$ ("Losses") – the three loss channels (whichever CX-loss model is active feeds $P_{\text{cx}}$) plus their sum $P_{\text{lost}} = P_{NB} - P_{\text{useful}}$.
- $Y_{\text{beam}}, Y_{\text{thermal}}, Y_{\text{bb}}, Y_{\text{full}}$ ("Neutrons") – the neutron rate split by reaction channel: beam-target (D-T + D-D), thermal-thermal (D-T + D-D), beam-beam (Sec. "Beam-beam fusion" above – a flat zero while that's off), and the full total ($Y_{\text{beam}} + Y_{\text{thermal}} + Y_{\text{bb}} = Y_{\text{full}}$ exactly). Named $Y$, not $R$, so it isn't confused with the $R = u_{\text{fast}}/U_{\text{th}}$ scan group.

## 1.25 Important notation

- $T_e$: electron temperature
- $T_i$: thermal-ion temperature
- $P_e$: beam power deposited into electrons
- $P_i$: beam power deposited into ions
- $L_e$, $L_i$: beam-power electron / ion heating fractions ($L_e + L_i = 1$)
- $P_{ie}$: electron-ion energy exchange power
- $P_{\text{ECRH}}$, $P_{\text{ICRH}}$: ECRH and ICRH auxiliary heating power
- $P_{\text{aux},e}$, $P_{\text{aux},i}$: auxiliary heating to electrons / ions
- $P_\alpha$, $P_{\alpha,e}$, $P_{\alpha,i}$: fusion alpha self-heating power (total, to electrons, to ions)

- $\tau_{E,e}$, $\tau_{E,i}$: electron / ion energy confinement times used in the balance
- $n_{\text{thermal}}$: thermal-ion density, $n_{\text{sum}}(n_{e0}, Z_{\text{eff}}) - n_{b0}$
- $n_D$, $n_T$: thermal deuterium and tritium densities
- $n_{b0}$: total fast-ion density
- $\rho_{Li}$: fast-ion toroidal-field Larmor radius at the injection energy
- $\Delta r$: passing drift-orbit radial width, $q_* \, \rho_{Li}$ (large-aspect model)
- $\rho_\theta$: fast-ion poloidal gyroradius, $(B_t/\bar{B}_p) \, \rho_{Li}$ (ST orbit models)
- $w_{\text{pass}}$, $w_{\text{ban}}$: ST circulating / banana orbit widths
- $f_t$, $f_c$: trapped / circulating particle fractions (arbitrary $A$)
- $q_a$: arbitrary-aspect-ratio edge safety factor (Uckan form)
- $f_{\text{orbit}}$: first-orbit loss fraction of captured beam power
- $P_{\text{orbit}}$: first-orbit loss power
- $P_{f,tot}$, $P_{f,th}$, $P_{f,b}$: total, thermal, and beam-target fusion power (D-T)
- $P_{f,DT}$, $P_{f,DD}$: D-T and D-D fusion power ($P_{f,tot} = P_{f,DT} + P_{f,DD}$)
- $R_n$: neutron rate (14 MeV D-T + 2.45 MeV D-D branch)
- $\langle T \rangle$, $T_0$: volume-averaged and on-axis temperature (profile-corrected 0-D)
- $p_n$, $p_{Te}$, $p_{Ti}$: density and electron / ion temperature profile exponents ($(1-\rho^2)^{2p}$)
- $\tau_{IE}$: effective electron-ion exchange time
- $p_{th}$, $p_{fast}$: thermal and fast-ion pressure
- $\beta_t$: toroidal beta
- $f_{\text{dil}}$: fast-ion dilution, $n_{b0}/(n_{\text{thermal}} + n_{b0})$
- $\tau_p^*$: effective particle confinement time (incl. recycling)
- $N_s$: thermal inventory of species $s$, $\langle n_s \rangle V$
- $S_{\text{beam}\to s}$, $S_{\text{ext},s}$, $R_{\text{burn},s}$: beam, external fuelling and burn-up particle rates

## 1.26 Appendix A: confinement-scaling $P$ definition – expected impact

This appendix expands on the caveat noted under "Confinement time" above. It is exploratory analysis, not a validated result: standalone scripts, outside the app and the solver, generated the three figures below (see `docs/Validation/tauE_scaling_Ploss_definition_scan.py` and `tauE_scaling_combined_comparison.py`). **Not yet acted on** – see the "confinement-scaling-p-loss-definition" note if revisiting this.

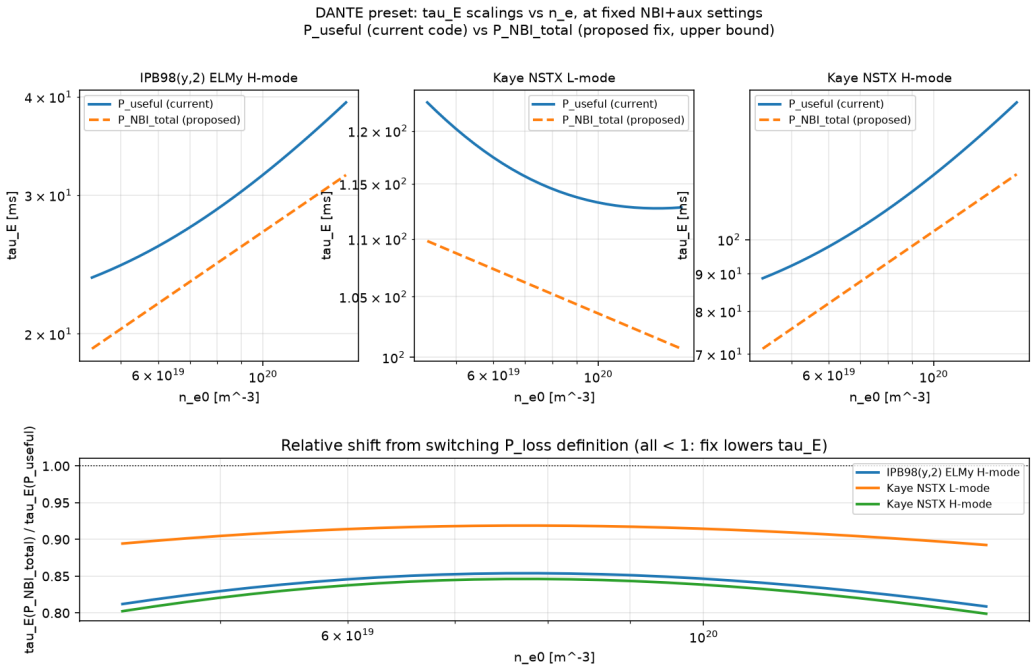
### 1.26.1 The question

IPB98(y,2) and the two Kaye NSTX scalings take a power argument, $P$. HI-Jass currently feeds them $P_{\text{useful}} + P_{\text{aux}}$ – NBI power *after* shine-through, first-orbit loss, and charge-exchange loss are all subtracted. The experimental databases these scalings were fit to use the coarser convention instead (absorbed power only, i.e. shine-through-corrected but *not* further reduced for orbit-loss/CX-loss of the already-absorbed fast-ion population). HI-Jass's convention therefore under-feeds $P$ into these negative-exponent scalings, which should **overestimate** $\tau_E$ whenever orbit-loss/CX-loss are significant – worst for compact/ST-like geometries such as this project's own DANTE default.

### 1.26.2 DANTE: how much would the fix change $\tau_E$?

For the DANTE preset, at fixed NBI/ECRH settings, switching from $P_{\text{useful}}$ to the raw total injected NBI power (an upper bound on the proposed fix – shine-through would still be subtracted in the real fix) lowers $\tau_E$ by **˜10-20%**, depending on the scaling and the density:

Orbit-loss + CX-loss account for ˜25-32% of DANTE's injected NBI power over this scan – that




gap is the direct cause of the shift above.

### 1.26.3 But the scaling choice itself matters far more

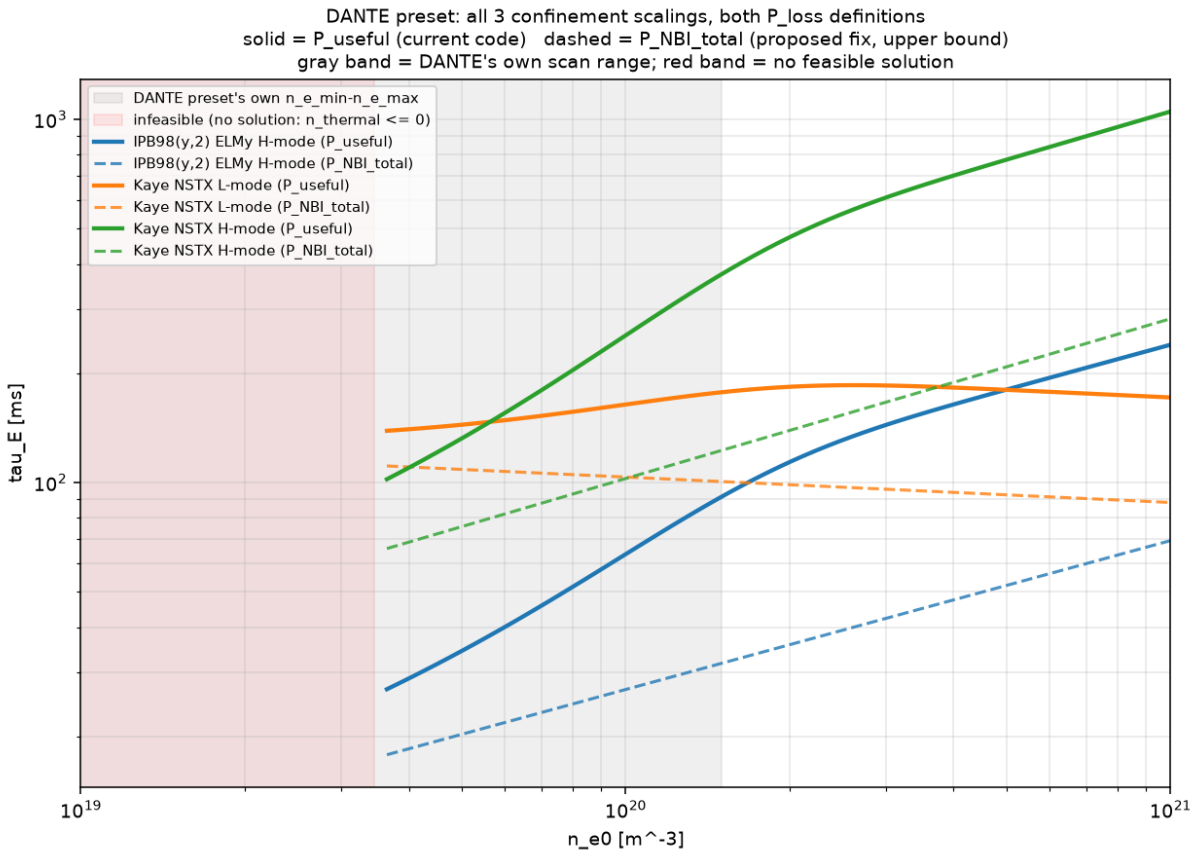
The same DANTE comparison, with all three scalings drawn on one axis, shows the **scaling-choice spread is ˜4-6x** across the density range – 20-30x larger than the $P$-definition effect above. Red shading marks densities where HI-Jass has no feasible solution at all for this preset (the model's thermal-ion density would go negative):

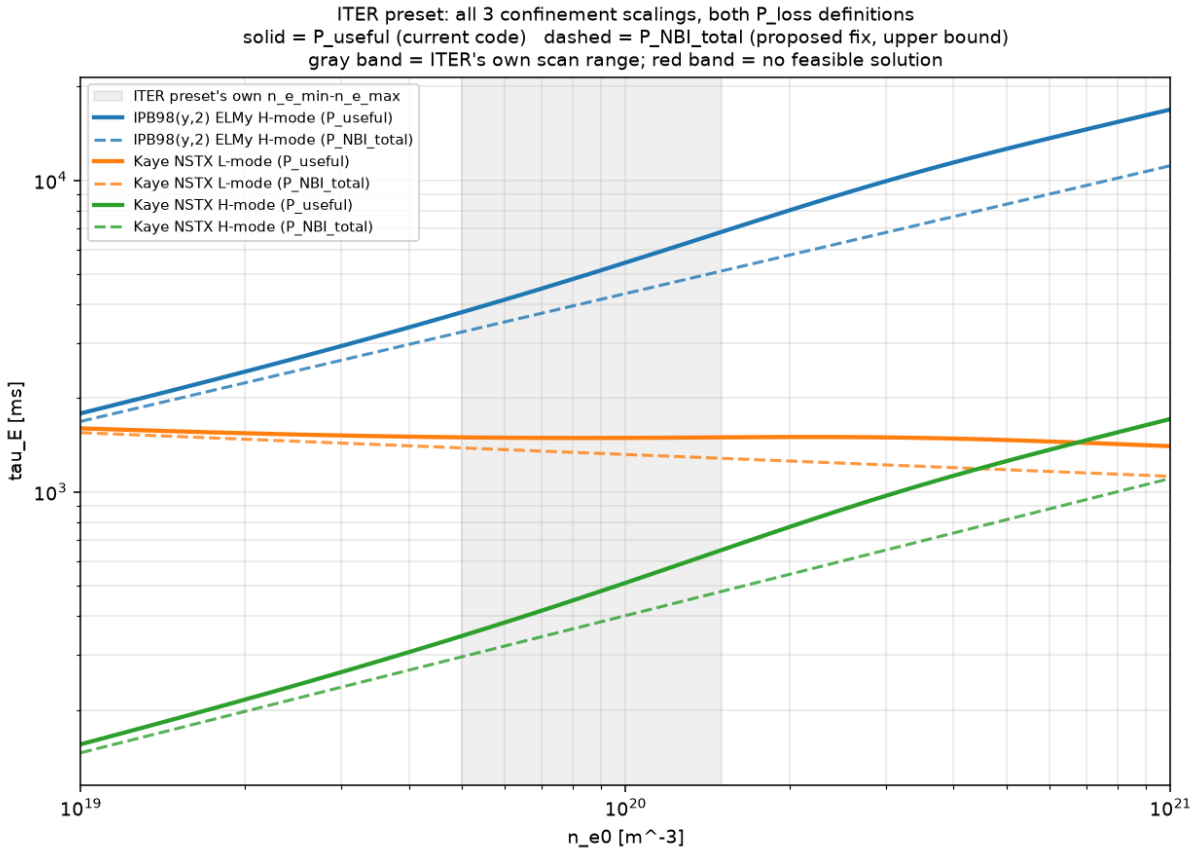
The same view for the **ITER** preset shows an even larger spread (˜10-12x), but with the *ranking reversed* – IPB98(y,2) now sits far above both Kaye curves, at multi-second $\tau_E$ values:

**Read this ITER case with caution, not as competing physics estimates.** Kaye's scalings were fit purely to NSTX data ($I_p \lesssim 1.5$ MA, $B_t \lesssim 0.5$ T); ITER's $I_p = 15$ MA / $B_t = 5.3$ T is a huge extrapolation *outside* that database, in the opposite direction from DANTE/NSTX-like devices. IPB98(y,2), by contrast, was fit including larger, more ITER-like devices, so it is the more trustworthy of the two here – the "spread" for ITER is mostly a statement about extrapolation validity, not a genuine disagreement in overlapping-validity physics.

### 1.26.4 Takeaway

Fixing the $P_{\text{loss}}$ definition (sec. "Known limitation" above) is a real, worthwhile correction (˜10-20% on $\tau_E$), but the choice of *which* scaling to use dominates the total uncertainty by an order of magnitude. Neither should be read as validated for a device far outside that scaling's own fitted dataset – check the Assumptions tab's low-aspect extrapolation flag, and prefer Kaye NSTX only for genuinely NSTX-like ($A \lesssim 1.6$, low $B_t$) geometries.


DANTE preset: all 3 confinement scalings, both P_loss definitions
solid = P_useful (current code)  dashed = P_NBI_total (proposed fix, upper bound)
gray band = DANTE's own scan range; red band = no feasible solution
DANTE preset's own n_e_min-n_e_max
infeasible (no solution: n_thermal <= 0)
IPB98(y,2) ELMy H-mode (P_useful)
IPB98(y,2) ELMy H-mode (P_NBI_total)
Kaye NSTX L-mode (P_useful)
Kaye NSTX L-mode (P_NBI_total)
Kaye NSTX H-mode (P_useful)
Kaye NSTX H-mode (P_NBI_total)
tau_E [ms]
n_e0 [m^-3]


ITER preset: all 3 confinement scalings, both P_loss definitions
solid = P_useful (current code)  dashed = P_NBI_total (proposed fix, upper bound)
gray band = ITER's own scan range; red band = no feasible solution
ITER preset's own n_e_min-n_e_max
IPB98(y,2) ELMy H-mode (P_useful)
IPB98(y,2) ELMy H-mode (P_NBI_total)
Kaye NSTX L-mode (P_useful)
Kaye NSTX L-mode (P_NBI_total)
Kaye NSTX H-mode (P_useful)
Kaye NSTX H-mode (P_NBI_total)
tau_E [ms]
n_e0 [m^-3]